

Coordinated Justice: The Politics of Prosecution and Policing*

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Abstract

Contemporary analyses of the politics of law have focused on policing and prosecution as distinct institutions in the criminal justice system. However, the implementation of law involves both institutions working in tandem to jointly produce the outcomes of the criminal justice bureaucracy. I develop a game-theoretic model of coordination between police and prosecutors in the implementation of criminal law. I model the interdependence of policing and prosecutorial charging decisions. The analysis reveals a strategic complementarity between these actions that generates multiple equilibria and institutional path dependence in enforcement regimes. I show that identical formal policies can sustain markedly different enforcement environments depending on coordination history, and reforms that fail to account for this interdependence may produce null or heterogeneous effects. The framework yields direct implications for the empirical analysis of the politics of criminal justice reform.

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When Larry Krasner took office as Philadelphia’s District Attorney in 2018, he changed which cases his office would pursue. He instructed his attorneys to decline a range of low-level charges, to divert others, and to treat most low-level retail theft as a summary offense. Like many other reform prosecutors, Krasner adopted policies that entailed a mix of blanket choices about which kinds of offenses to prioritize and other less formal policies about how much evidence would be necessary to prosecute a crime. Over the ensuing years the changes generated sustained controversy and tension with the Philadelphia Police Department. Yet despite the shift in charging priorities, arrest and conviction rates across the city did not change in a marked way. In March 2025, Krasner’s office partially reversed course, and in November 2025 he was re-elected to a third term.

A similar agenda unfolded in San Francisco. After taking office as District Attorney in 2020, Chesa Boudin similarly narrowed the set of cases his office would pursue, declining to prosecute quality-of-life offenses such as public camping, solicitation, and public urination. As in Philadelphia, these changes that targeted specific kinds of offenses provoked backlash from police officials. However, unlike in Philadelphia, they did coincide with visible changes in enforcement patterns and became a focal point of political contestation. In 2022, Boudin was removed from office in a recall election.

These cases present a puzzle. Prominent reform prosecutors in two large cities adopted broadly similar reforms, yet the system-level consequences diverged sharply. It is tempting to conclude that one reform succeeded and the other failed. I argue instead that the reforms differed less in their content than in how police and prosecutors had been coordinating beforehand, and that this difference has direct consequences for what we can infer from the data. Existing empirical designs treat changes in charging or conviction rates as direct reflections of prosecutorial policy. But if police adjust their behavior in anticipation of how prosecutors will treat cases, those rates conflate the selection of cases into the prosecutorial pipeline with outcomes conditional on entry, and standard designs may misattribute a reform’s effects or miss them entirely. Criminal justice enforcement is the joint product of strategic interaction between prosecutors and police, and much of the politics of criminal justice resides in that interaction.

I develop a theory of criminal justice enforcement as a coordination problem between police and prosecutors. Police identify suspects, collect evidence, and assemble case files, whereas prosecutors have responsibility for and discretion over formal charging decisions, including whether to charge a suspect in the first place. And, they have multiple tools in that process to influence how law enforcement operates. The Krasner and Boudin reforms operated by setting prosecutorial priorities, defining which offenses their offices would pursue. A second, less visible but also consequential, tool is discretion over the evidentiary standard that governs how much evidence a case must contain before the office will pursue it. A core function for prosecutors is to screen a case's evidentiary support via, among other things, corroborating evidence and subjecting decisions to supervisory approval. In this paper, I focus on the evidentiary standard and on how it interacts with the mix of offenses that police bring forward. I model the process by which police allocate effort across areas of law enforcement, and invest in building case files, in anticipation of how prosecutors will treat what they produce. Prosecutors, in turn, set their standards in the shadow of electoral accountability. Thus, prosecutorial standards and police evidence investment serve as strategic complements, and criminal justice enforcement has the structure of a coordination game. Path dependence, equilibrium multiplicity, and reform fragility are then properties of the institutional system itself, not artifacts of measurement or implementation failure.

I highlight three key findings. First, higher charging standards need not reduce conviction rates, because they can affect the selection of cases into the set that prosecutors consider charging in the first place. As a consequence, it need not be that higher prosecutorial standards produce fewer convictions. Second, prosecutorial standards shape police effort and priorities, and they can generate multiple enforcement equilibria that differ in both evidence production and charging behavior. We can therefore observe different enforcement patterns and environments under identical formal policies, as a result of coordination between police and prosecutors. Third, due to equilibrium persistence, reforms may change policy without altering the underlying coordination, and so may bring about no observable change in aggregate outcomes, despite altering the dynamics of the police-prosecutor interaction.

Integrating prosecutors and police as coordinated actors driving law enforcement has direct implications for a large and growing body of empirical scholarship. Research on prosecutorial reform increasingly asks whether police respond to changes in charging policy, and it often treats that response as a threat to identification to be detected and controlled (e.g., Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023), even while a wide range of work points out that the two functions are deeply interdependent (e.g., Richman 2003, Sklansky 2018). However, we do not yet have a rigorous framework for evaluating or understanding the nature of the interdependence (or its implications for our inferences). Similarly, most research on prosecutorial discretion typically studies how prosecutors handle cases without accounting for the political forces that shape the pool of cases they process in the first place (e.g., Gordon and Huber 2002, Landes 1971), while the policing literature does not integrate endogenous downstream decisions (i.e., prosecutors) as a part of how police do their work in the first place (e.g., Wilson 1968). But, when we put these two institutions together and treat them as part of a single system, then the strategic interdependence of police and prosecutors has implications for what we can infer from aggregate outcomes such as arrest, charging, and conviction rates (e.g., Agan et al. 2025).

The framework speaks to a growing empirical literature on the effects of prosecutorial reform (e.g., Agan, Doleac and Harvey 2023, Agan et al. 2025, Bandyopadhyay and McCannon 2014, Dobbie, Goldin and Yang 2018, Goncalves and Mello Forthcoming, Pfaff 2017, Soliman 2022) and on police behavior (e.g., Ba et al. 2021, Clark 2024, Eckhouse 2022, Mummolo 2018). The insight here in particular is the articulation of a formal account of how policing and prosecution are jointly determined and what that joint determination implies. I show that treating criminal justice as an equilibrium outcome of coordinated behavior clarifies a variety of potential empirical evaluations. It helps explain how to interpret out empirical findings and shows that our analytic approaches need to account for the coordination dynamics that produce them.

1 Prosecutors and Police

Recent literature has developed two increasingly sophisticated but largely separate research programs on the central actors in American criminal law enforcement. One studies prosecutors: their unilateral charging authority, the role of electoral accountability in shaping their behavior, and the consequences of their discretion for who is prosecuted and on what terms. The other studies police: their organizational incentives, resource allocation decisions, and the determinants of enforcement patterns. Each of these lines of research has generated important insights, but each also takes as given features of the institutional environment that are, in fact, endogenous to the other actor's choices. The result is that our theoretical understanding of criminal justice politics is built on a foundation that treats as separate what is, in practice, a single interdependent system.

Prosecutors exercise unilateral formal authority over charging decisions and are directly accountable to the public; police are far less electorally accountable. The literature on prosecutorial discretion emphasizes the consequences of that discretion for who is prosecuted and on what terms (e.g., Barkow 2019, Davis 2007, Gordon and Huber 2009, Landes 1971), and treats electoral accountability as the prosecutor's primary constraint (e.g., Gordon and Huber 2002). However, these analyses typically assume the distribution of cases a prosecutor handles is exogenously shaped. That assumption is consequential and potentially problematic to the extent that police anticipate and respond to prosecutorial standards, making the pool of cases a prosecutor faces is itself a strategic object.

Prosecutors depend heavily on police, who determine which cases are investigated, what evidence is collected, and how cases are packaged for prosecution (Goldstein 1963, Greenwood, Chaiken and Petersilia 1977, Lee 2020). Prosecutors, unlike police, are politically (typically electorally) accountable for the state of public safety. Via elections prosecutors are subject to reputational pressure, media scrutiny, career concerns, and the latent threat of a challenge which can all affect how they craft policy and make decisions about charging criminal cases. I therefore treat elections as the most visible instance of a more general accountability pressure,

and nothing in the analysis turns on that pressure operating through close races.¹

On the other hand, research in police behavior and organizations typically assumes that a prosecutor’s downstream charging practices standards are fixed, exogenous features, rather than strategic responses to policing practices (e.g., Goldstein 1963, Goncalves and Mello Forthcoming, Soliman 2022, Wilson 1968). That is consequential because if a policing agency finds it costly to pursue cases that the prosecutor may set aside due to high evidentiary expectations, the agency may simply redirect effort toward categories where it can more easily or reliably produce prosecutable evidence that meets the prosecutorial bar. Similarly, if a prosecutor adopts a more permissive standard, then then return to investing in investigative effort declines, and so police may instead prioritize arrest volume over case quality (Forst and Brosi 1977, Pfaff 2017). In this sense, prosecutors use policies and standards as mechanisms of institutional delegation that, in practice, shapes the composition and quality of cases that enter the system without prosecutors exerting direct authority over policing (Sklansky 2018).

While the institutional arrangement between police and prosecutors channels formal authority into practical influence, both the police and the prosecutors still operate in the shadow of a court, where any conviction must ultimately take place. The risk of acquittal and the reputational cost of failed prosecutions constrain the cases a prosecutor can viably charge, making the quality of police-generated evidence a matter of political salience rather than administrative preference (Bandyopadhyay and McCannon 2014, Bibas 2004, Boylan and Long 2005, Eisenstein and Jacob 1977, Nardulli, Eisenstein and Flemming 1988). One question I address below is how the anticipation of the judicial process shapes the interaction between police and prosecutors and, in turn, complicates standard agency-based models of law enforcement. Further, the model I build helps illuminate why political pressure on prosecutors need not translate mechanically into charging outcomes. Because police reallocate effort in anticipation of screening, the mapping from public preferences to enforcement is mediated at every stage by the broader

¹Of course, most American prosecutors That accountability is usually described in electoral terms, but most chief prosecutors run unopposed, and local electoral competition is, on its own, weak (Hessick and Morse 2019, Rogers 2023, Wright 2009). What the model requires is not contested elections but political accountability broadly.

institutional setting that makes up criminal law enforcement (Gordon and Huber 2002, Huber and Gordon 2004, Wilson 1968).

To be sure, recent work has begun to relax these assumptions, examining how police respond to prosecutorial policy (e.g., Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023), but what remains missing is an account of their joint determination. However, the theoretical literature typically take the set of cases a prosecutor handles as given and focused on optimization within it, including investigative resource allocation and plea bargaining (Baker and Mezzetti 2001), charging under case-level uncertainty (Bjerk 2007), and the objectives prosecutors pursue among federal cases (Glaeser, Kessler and Morrison Piehl 2000), while work on optimal law enforcement (Shavell 1993) models enforcement intensity but not the strategic production of evidence by a separate agency with its own accountability. Treating prosecution and policing as a joint coordination problem uncovers important nuance, including equilibrium multiplicity and institutional path dependence that arises from the strategic complementarity between an electorally accountable principal and a semi-autonomous evidence-producing agency.

2 The Model

I model an institutional setting in which a police agency produces evidence and an elected prosecutor exercises charging discretion under electoral accountability. The model isolates the strategic nature of prosecutorial charging decisions responding to how police produce information and, in turn, how prosecutorial evidentiary standards shape police priorities and investment incentives across different areas of law enforcement. Conceptually, the game is played repeatedly across a large number of cases, but for expositional ease I treat it as a single-period, static model. In the analysis below, I treat equilibrium behavior as generating outcomes over a portfolio of cases.

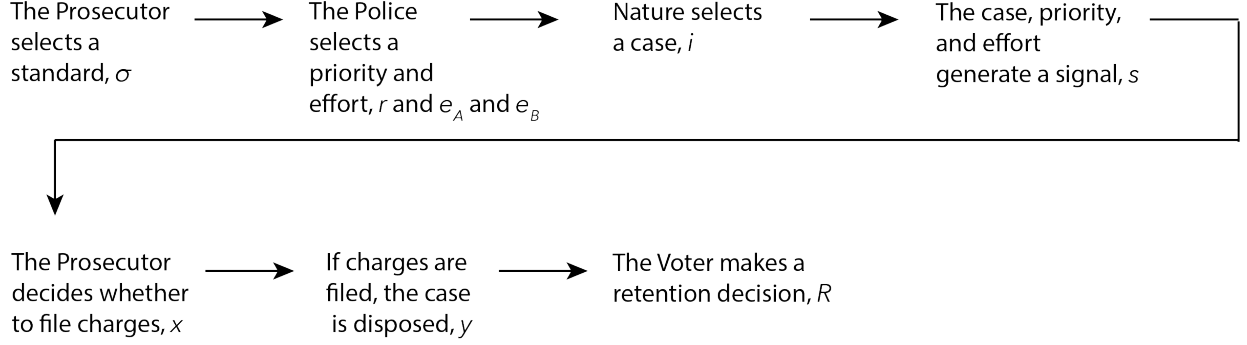


Figure 1: Timing of events in the model

2.1 Players and sequence of Play

There are two strategic players: an elected prosecutor P and a police agency L . I treat the agency as a single player who moves in two stages. We can conceive of the agency, though, as acting through a commissioner, who sets enforcement priorities, and line officers, who choose investigative effort, under the assumption the two share the agency's objective. There are also two non-strategic actors, a trial court J and a voter V .

The game unfolds in seven steps, summarized in Figure 1:

1. First, the prosecutor commits to an evidentiary standard $\sigma \in \{\text{strict}, \text{lenient}\}$, publicly observed by the agency, which constrains her subsequent charging standards.
2. Second, the commissioner observes σ and chooses a priority parameter $r \in [0, 1]$ that determines the distribution of cases across two types of cases, with $\Pr(i = A) = r$ and $\Pr(i = B) = 1 - r$.
3. Third, the line officers observe σ and r and then choose evidence investments $e_A, e_B \in [0, \bar{e}]$ subject to a total effort budget $re_A + (1 - r)e_B \leq \bar{B}$.
4. Fourth, Nature draws the case type i and an unobserved state $\theta \in \{0, 1\}$ with $\Pr(\theta = 1 \mid i) = \pi_i$, where $\theta = 1$ denotes a guilty defendant and $\theta = 0$ an innocent one; whether a guilty defendant is convicted is endogenous, determined below by the evidence and the court. The case file produces a binary signal $s \in \{H, L\}$ with $\Pr(s = H \mid \theta = 1, i, e_i) = \alpha_i(e_i)$ and $\Pr(s = H \mid \theta = 0, i, e_i) = \beta_i(e_i)$, where $\alpha'_i(e) > \beta'_i(e) \geq 0$: more effort raises

the chance of a strong file, and more so when the defendant is guilty.

5. Fifth, the prosecutor observes (σ, i, s) but not θ or the police choices (r, e_A, e_B) and selects a charging rule $x : \{A, B\} \times \{H, L\} \rightarrow \{0, 1\}$, given the constraints imposed by her choice of σ , where $\mathcal{X}_{\text{strict}} = \{x : x(i, H) = 1, x(i, L) = 0\}$ and $\mathcal{X}_{\text{lenient}} = \{x : x(i, H) = 1, x(i, L) \in \{0, 1\}\}$.
6. Sixth, if $x = 1$ the court adjudicates the case, yielding $y \in \{0, 1\}$ ($y = 1$ a conviction), with $\Pr(y = 1 \mid x = 1, \theta, s) = q(\theta, s)$ more likely when the defendant is guilty and the signal is high; q is exogenous and reflects the judicial environment. If $x = 0$ there is no adjudication and I set $y = 0$.
7. Finally, the voter observes coarse aggregate measures of performance over the period's portfolio (convictions, failed prosecutions, declinations) and the announced standard σ , then decides whether to retain the prosecutor, $D \in \{0, 1\}$.

2.2 Elections and Aggregation

While I model the game at the level of a single case, it is most natural to interpret it as describing a representative case drawn from a larger portfolio of cases that are handled over an election period. In particular, we should not think of the Voter as conditioning reelection on any one case outcome in isolation. Instead, she observes coarse summary measures of office performance (e.g., charging rates, conviction rates, and salient failures) generated by the portfolio.

Let $\Gamma(\cdot)$ map aggregate summary measures of performance into the prosecutor's probability of retention. For expositional convenience, I write Γ as a function of the representative case's contribution to convictions, acquittals, and declinations. I hold the distribution of outcomes in the remaining cases fixed and average over it in Γ . The prosecutor's retention probability is then

$$\Pr(D = 1 \mid x, y, \sigma) = \Gamma(y, x(1 - y), 1 - x, \sigma).$$

With this technology, it is direct to capture the notion that electoral accountability responds to

aggregate prosecutorial outcomes, including cases not prosecuted even while modeling the game as the resolution of a single case. I impose minimal structure on $\Gamma(\cdot)$, so that voters may evaluate performance using either conditional outcomes (e.g., conviction rates among prosecuted cases) or unconditional outcomes (e.g., conviction or declination rates across all cases). A broad class of evaluation rules can support the inferences. Because Γ enters only in reduced form, elections serve only to shape prosecutorial choices, not to determine the underlying prosecutor type.²

The corresponding population rates are

$$\bar{y} \equiv \mathbb{E}[y], \quad \overline{x(1-y)} \equiv \mathbb{E}[x(1-y)], \quad \overline{(1-x)} \equiv \mathbb{E}[1-x].$$

These expectations are taken over the election-period distribution of cases induced by agency choices (r, e_A, e_B) . The priority choice r determines the distribution of cases across categories, while evidence investment determines the distribution of signals within each category. As a result, charging and conviction rates are a function of both prosecutorial decisions and the cases police produce. Changes in either rate can be due to a change in how the prosecutor treats a given file or a change in the composition and evidentiary quality of the files that reach the prosecutor. I return to this distinction below.

2.3 Preferences

Prosecutor. I assume the prosecutor values convictions, dislikes losing charged cases, dislikes convicting the innocent, and values holding office. The prosecutor's per-case payoff is given by

$$U_P = b \cdot y - a \cdot x(1-y) - \lambda \cdot y(1-\theta) + \rho \cdot \Pr(D=1 \mid \text{history}),$$

where $b > 0$ captures the professional and reputational benefit of a conviction, $a > 0$ measures the reputational cost of a failed prosecution (especially a salient acquittal), $\lambda > 0$ is the normative cost of convicting an innocent defendant (the event $y = 1$ while $\theta = 0$), and $\rho > 0$ is the value of retention. The parameters b and a measure intrinsic interests that go beyond simply

²Note also this implies a limit on voter sophistication. A strategic voter who updated on the composition of outcomes would induce some evaluation rule.

holding office, including professional standing, office morale, and credibility with police and the courts. Electoral accountability affects the prosecutor’s utility through Γ . I assume that the normative costs a prosecutor faces come from a wrongful conviction. Declining to charge gives an intrinsic payoff of zero, while the electoral consequences arise through retention.

Police agency. The police agency values having cases charged (and, optionally, convicted) and incurs costs of (i) allocating enforcement attention across categories and (ii) producing evidentiary files. The agency values charges and convictions as enforcement output rather than conditioning on guilt, consistent with the prominence of arrest and clearance statistics as measures of police productivity. Let $r \in [0, 1]$ denote the induced share of cases in category A . The police agency’s expected payoff is

$$\begin{aligned} \mathbb{E}[U_L] = & r \left(v_A \Pr(x = 1 \mid A) + w_A \Pr(y = 1 \mid A) - c(e_A) \right) \\ & + (1 - r) \left(v_B \Pr(x = 1 \mid B) + w_B \Pr(y = 1 \mid B) - c(e_B) \right) - C(r), \end{aligned}$$

where c is increasing and convex and C is convex with an interior minimum. Here $c(e_i)$ is the per-case resource cost of generating evidentiary quality in category i . $C(r)$ measures the organizational cost of *concentrating* enforcement—shifting patrol assignments, standing up or disbanding detective units, and the political frictions of reallocation.

As noted above, L comprises both a commissioner and line officers. The commissioner chooses the allocation r , anticipating officer effort; line officers then choose (e_A, e_B) given r and the budget constraint, $\bar{B}$. The effort budget means there is an interaction between the allocation and effort choices. Because investigative capacity is shared, the effort officers can devote to a case depends on how many cases the commissioner routes into each category. That plays a distinct role from the reallocation cost, $C(r)$, represents the idea that concentrating resources in either category is costly.

3 Analysis

To analyze the model, I first describe the core logic underlying the prosecutor and police agency's decision calculus. I then turn to an equilibrium analysis and the behavioral properties that arise in equilibrium. Section 4 then builds on this characterization to examine coordination, equilibrium selection, and the implications of multiplicity for prosecutorial reform and analysis of the effects of reform.

3.1 The Prosecutor's Charging Condition

The prosecutor's evidentiary standard, σ , is chosen as an ex ante commitment device and applies across cases rather than being chosen case by case. To be incentive compatible, then, a standard must be optimal given the prosecutor's expectations about the portfolio of cases generated by the police response to the standard. Let $\mu_i(s; \sigma)$ denote the prosecutor's posterior belief that a defendant from activity type i is guilty, given the evidentiary signal s and the prosecutor's chosen standard σ . Formally,

$$\mu_i(s; \sigma) \equiv \Pr(\theta = 1 \mid i, s, \sigma) = \frac{\pi_i \Pr(s \mid \theta = 1, i, \sigma)}{\pi_i \Pr(s \mid \theta = 1, i, \sigma) + (1 - \pi_i) \Pr(s \mid \theta = 0, i, \sigma)},$$

where $\pi_i \equiv \Pr(\theta = 1 \mid i)$ is the prior probability that a case of type i involves a guilty defendant. The police agency's equilibrium investment, $e_i(\sigma)$, is determined in equilibrium by the evidentiary standard σ , which is publicly observed. Because the prosecutor does not observe $e_i(\sigma)$, her equilibrium charging decisions are a function of σ , and the induced distribution of signals about cases.

Next, define the per-case *electoral marginal incentive* from charging as

$$\Delta_E(i, s; \sigma) \equiv \rho(\mathbb{E}[\Gamma(\cdot) \mid x = 1, i, s, \sigma] - \mathbb{E}[\Gamma(\cdot) \mid x = 0, i, s, \sigma]).$$

That incentive captures how much more electoral return the prosecutor gets from charging a given case, as opposed to not charging it. It incorporates how the electoral environment rewards or penalizes the act of filing charges, conditional on the evidentiary class of the case and the

prosecutor's announced standard.

These objects allow us to characterize the prosecutor's optimal charging condition. Both charging and declination are feasible when the prosecutor faces a weak file under a lenient standard. Thus, the prosecutor's charging condition is given by:

$$1 \in BR_P(i, s; \sigma) \iff b \cdot \Pr(y = 1 \mid x = 1, i, s) - a \cdot \Pr(y = 0 \mid x = 1, i, s) \\ - \lambda \cdot \Pr(\theta = 0 \mid i, s, \sigma) q(0, s) + \Delta_E(i, s; \sigma) \geq 0,$$

where $\Pr(y = 1 \mid x = 1, i, s) = \mu_i(s; \sigma) q(1, s) + (1 - \mu_i(s; \sigma)) q(0, s)$.

This expression of the prosecutor's charging reveals the central tradeoff at the heart of the model. The prosecutor weighs the substantive and electoral benefits of a potential conviction against three countervailing forces. Those include the reputational cost of an acquittal, the normative cost of convicting an innocent defendant, and the electoral consequences of filing charges. Recall that her expectations about case quality depend on the police response induced by σ . As a result, her charging calculus itself determines what standard (lenient or strict) can be supported in equilibrium. But it also determines how police investment must adjust to make those standards incentive compatible. As I show in the equilibrium analysis below, these constraints play a central role in shaping police priorities and the existence of distinct equilibria.

3.2 Police priority allocation and effort investments

Consider next the police agency's incentives. Working backwards, line officers choose effort given the allocation, r , and the commissioner then chooses the allocation anticipating that effort. To see how these forces interact, let

$$\Pi_i(\sigma) \equiv v_i \Pr(x=1 \mid i, \sigma) + w_i \Pr(y=1 \mid i, \sigma)$$

denote the police agency's expected per-case gross return from a case in category i under standard σ , evaluated at the equilibrium effort induced by σ .

Officer effort. Given (σ, r) and the budget constraint $\bar{B}$, officers allocate investigative effort across the two categories of cases. An interior effort must satisfy

$$v_i \frac{\partial \Pr(x = 1 \mid i)}{\partial e_i} + w_i \frac{\partial \Pr(y = 1 \mid i)}{\partial e_i} - c'(e_i) = \mu, \quad i \in \{A, B\},$$

where $\mu \geq 0$ is the marginal value of additional investigative capacity. If capacity is unconstrained, then $\mu = 0$. When the budget binds, officers allocate effort until the net marginal return equals μ in both categories.

The two terms on the left capture the two ways that evidence quality affect L 's utility. First, it raises the probability that the prosecutor files charges (weighted by v_i). Second, it increases the probability of eventual conviction (weighted by w_i). How much L values each depends on the charging standard, σ . Under a strict standard, charging is sensitive to evidentiary strength, so officers value both terms. Under a lenient standard, though, the prosecutor charges the case regardless of signal quality, and so $\Pr(x=1 \mid i)$ is invariant to effort. Investigation only benefits the police via its effect on the likelihood of conviction. Further, because strict screening activates the charging channel, it raises the demand for effort. As a result budget is then more likely to bind, and so the shadow value of capacity tends to be higher under strict screening.³

Commissioner priorities. The commissioner chooses r anticipating officer effort. Assuming the agency's optimized payoff function is differentiable, we can calculate the necessary condition for an interior solution to the choice of priorities via the envelope theorem:

$$\begin{aligned} & \left[v_A \Pr(x = 1 \mid A, \sigma) + w_A \Pr(y = 1 \mid A, \sigma) - c(e_A) \right] \\ & - \left[v_B \Pr(x = 1 \mid B, \sigma) + w_B \Pr(y = 1 \mid B, \sigma) - c(e_B) \right] - \mu(e_A - e_B) = C'(r). \end{aligned} \quad (1)$$

where $\mu \geq 0$ is the Lagrange multiplier on the investigative-capacity constraint, and therefore measures the marginal value of additional capacity. The first two terms on the left-hand side capture the agency's expected net return from cases in categories A and B , while the last term accounts for the investigative capacity each category requires. What we see here is that the

³This is a tendency rather than a general inequality.

commissioner will shift cases toward category A until its relative return, net of its demand on capacity, equals the marginal cost of reallocation. Because the prosecutor's standard changes these returns and the effort required to produce them, it affects both the evidence police produce and the categories they prioritize.

Before turning to equilibrium behavior, it will be useful to characterize optimal policing behavior. When the police know that the prosecutor will charge a given case, investigation benefits police only through its effect on convictions. However, when the prosecutor will only charge a case that has a strong file, investing in investigation also increases the chance that a case will be prosecuted. But, because there is a finite budget to be allocated across the two categories of cases, the agency's choice of which cases to pursue also depends on the resources needed to investigate them.

Lemma 1 (Police choices given charging) *Fix a charging rule with $x(i, H) = 1$ in both categories. Suppose each category's expected return net of investigation costs is concave in effort, and C is differentiable. A feasible police choice (r, e_A, e_B) with $0 < r < 1$ is optimal given this charging rule if and only if it satisfies equation (1) and the following conditions for a common $\mu \geq 0$, the multiplier on the effort budget.*

At an interior effort choice, an unscreened category ($x(i, L) = 1$) satisfies

$$w_i \left[\pi_i \alpha'_i(e_i) \{q(1, H) - q(1, L)\} + (1 - \pi_i) \beta'_i(e_i) \{q(0, H) - q(0, L)\} \right] = c'(e_i) + \mu. \quad (2)$$

A screened category ($x(i, L) = 0$) instead satisfies

$$v_i [\pi_i \alpha'_i(e_i) + (1 - \pi_i) \beta'_i(e_i)] + w_i [\pi_i \alpha'_i(e_i) q(1, H) + (1 - \pi_i) \beta'_i(e_i) q(0, H)] = c'(e_i) + \mu. \quad (3)$$

At $e_i = 0$, replace the equality in the applicable condition by $\leq$; at $e_i = \bar{e}$, replace it by $\geq$. The budget conditions are

$$re_A + (1 - r)e_B \leq \bar{B}, \quad \mu[\bar{B} - re_A - (1 - r)e_B] = 0.$$

3.3 Equilibria

To solve the model, I characterize pure-strategy Perfect Bayesian Equilibria (PBE). It will be useful throughout the equilibrium analysis to define the *net marginal gain from charging* a case

of type i with signal s under standard σ :

$$\Phi(i, s; \sigma) \equiv b \cdot \Pr(y=1 \mid x=1, i, s) - a \cdot \Pr(y=0 \mid x=1, i, s) - \lambda(1 - \mu_i(s; \sigma))q(0, s) + \Delta_E(i, s; \sigma). \quad (4)$$

By assumption, under any standard, the prosecutor charges when the case file is strong. That assumption imposes constraints on the model parameters to ensure that $\Phi(i, H; \sigma) > 0$, for $i \in \{A, B\}$. What equilibrium can be supported therefore depends on (i) the sign of $\Phi(i, L; \sigma)$ —the net gain from charging a weak-evidence case when such charging is admissible—and (ii) the prosecutor’s (*ex ante*) expectations about payoffs under each standard. For the latter, let $V_P(\sigma) \equiv \mathbb{E}[U_P \mid \sigma, r^*(\sigma), e^*(\sigma)]$ denote the prosecutor’s expected payoff under standard σ , evaluated at a specified continuation equilibrium, including the police agency’s best response ($r^*(\sigma), e^*(\sigma)$). A continuation specifies charging decisions and beliefs as well as police behavior after that standard. For σ to be an equilibrium strategy, then, it must be *ex ante* optimal for the prosecutor—i.e., $V_P(\sigma) \geq V_P(\sigma')$ for $\sigma' \neq \sigma$. Both values refer to the continuations prescribed in the same PBE. I use the interior allocations and full signal support specified in Appendix A-2.1 throughout the equilibrium analysis.

The sign of $\Phi(i, L; \sigma)$ and the prosecutor’s expectations give rise to three distinct types of equilibria. The unscreened equilibrium arises when electoral pressure dominates evidentiary discipline; the screened equilibrium arises when costs of weak prosecutions dominate electoral pressure; and the category-dependent equilibrium arises when cross-domain heterogeneity creates evidentiary discipline only in some domains.

3.3.1 Unscreened Charging Equilibrium

The first type of equilibrium arises when electoral incentives reward visible enforcement (i.e., charging) over evidentiary discipline (i.e., conviction). The prosecutor adopts a lenient standard and charges all cases regardless of evidentiary strength. Anticipating that evidence quality does not affect charging decisions, police underinvest in diagnostic evidence and concentrate enforcement on high-volume categories. This equilibrium involves elevated charging rates, weak

responsiveness of charging to case strength, low evidence quality, and a case mix tilted toward categories where large numbers of cases can be generated at low investigative cost.

Proposition 1 (Unscreened charging equilibrium) *Suppose that (a) $\Phi(i, L; \text{lenient}) \geq 0$ for all $i \in \{A, B\}$ and (b) $V_P(\text{lenient}) \geq V_P(\text{strict})$. Then there exists a PBE in which:*

1. *the prosecutor adopts $\sigma = \text{lenient}$;*
2. *charging is unscreened on evidentiary strength: $x(i, H) = x(i, L) = 1$ for all i ;*
3. *police choices (r, e_A, e_B) satisfy the conditions in Lemma 1, using the unscreened effort condition (2) in both categories.*

The two conditions necessary to support this equilibrium ensure that the prosecutor is willing to accept the consequences of charging weak files for both types of cases and that the prosecutor cannot do better with a strict standard, given the changes in police behavior that standard would induce. Specifically, Condition (a) requires that for each case type, the electoral benefit of charging a weak file exceeds the combined reputational cost of acquittal and normative cost of a wrongful conviction. Condition (b) requires that the prosecutor prefers the portfolio of outcomes under lenient screening—given the police behavior it induces—to those attainable under strict screening with the higher evidence investment strict would elicit. Because charging is not affected by evidence quality, the agency has little reason to expend effort. We should expect, then, that police will not work to collect high-quality diagnostic evidence, and evidentiary signals will be of limited informative value. Section 4 shows why this low-investment environment can be self-sustaining.

3.3.2 Screened Charging Equilibrium

A second type of equilibrium arises when failed prosecutions are politically salient and wrongful charging carries meaningful costs. That incentivizes prosecutors to carefully scrutinize a file to ensure a high likelihood of conviction. The prosecutor commits to a strict standard, filing charges only when the case file is strong. The police agency responds by investing in evidence quality, concentrating effort on cases where investment can reliably produce strong

files. Empirically, this equilibrium produces lower charging rates, strong sensitivity of charging to evidentiary strength, higher evidence quality, and a shift toward categories with high investigative productivity.

Proposition 2 (Screened charging equilibrium) *Suppose $V_P(\text{strict}) > V_P(\text{lenient})$. Then there exists a PBE in which:*

1. *the prosecutor adopts $\sigma = \text{strict}$;*
2. *charging is screened on evidentiary strength: $x(i, H) = 1$ and $x(i, L) = 0$ for all i ;*
3. *police choices (r, e_A, e_B) satisfy the conditions in Lemma 1, using the screened effort condition (3) in both categories.*

Strict screening gives rise to another incentive for the police. Investment in effort increases the probability that any given case will meet the charging threshold, which is not a consideration in the unscreened charging equilibrium. The police agency therefore reallocates both investigative effort and case selection toward categories where evidence production is most effective. (Thus, as noted above, optimal effort investment includes the effect it will have on charging.) To support this equilibrium, there must be some combination of relatively larger reputational cost of acquittals (a), high costs to wrongful-conviction cost (λ), voter sensitivity to failed cases (through Γ), and a courtroom environment ($q()$) that imposes meaningful scrutiny on weak evidence. That is, prosecutors face a tradeoff between inducing police effort but risking not charging cases because the agency did not produce a strong enough file, or risking losing cases because they charge cases with weak evidentiary bases. In effect, the agency shifts from volume-maximizing enforcement toward diagnostic investigation, yielding a pattern of fewer, stronger cases.

3.3.3 Category-Dependent Equilibrium

A final type of equilibrium I consider captures selective toughness. In this equilibrium prosecutorial screening varies across policing domains even though preferences and formal authority are uniform. This equilibrium arises when categories differ in baseline convictability, courtroom scrutiny, or the productivity of evidence investment. This equilibrium has heterogeneous

charging thresholds across categories, with evidence quality and sensitivity to case strength improving only in domains where prosecutorial screening is effective.

Proposition 3 (Selectively screened equilibrium) *Suppose, without loss of generality, that $\Phi(A, L; \text{lenient}) \geq 0$ and $\Phi(B, L; \text{lenient}) < 0$, and that $V_P(\text{lenient}) \geq V_P(\text{strict})$. Then there exists a PBE in which:*

1. *the prosecutor adopts $\sigma = \text{lenient}$;*
2. *charging is unscreened in category A and screened in B: $x(A, H) = x(A, L) = 1$, $x(B, H) = 1$, $x(B, L) = 0$;*
3. *police choices (r, e_A, e_B) satisfy the conditions in Lemma 1, using the unscreened effort condition (2) in category A and the screened effort condition (3) in category B.*

In this equilibrium, the prosecutor's screening standard induces the police to treat the different categories of crime differently. For cases treated with the permissive standard, investing in evidence affects convictions but has no effect on whether charges are filed. In the screened category, that investment affects both conviction and the likelihood of meeting the charging standard. As a result, the policing priorities depend on comparing the returns to pursuing cases from each category, given that one can have an effect on charging, and the effort budget. The result is the possibility of differences in both case allocation and investigative effort.

3.4 Equilibrium Selection and Multiplicity

Unsurprisingly, under some parameter conditions, there are multiple equilibria to the game. In particular, given a lenient standard, there exists a subtle form of multiplicity. Under $\sigma = \text{lenient}$, the prosecutor's commitment permits *but does not require* charging on weak files ($s = L$), and the payoff to charging weak files depends on the evidentiary environment induced by police investment. Equilibrium multiplicity can arise as a result of that residual discretion.

To make this interaction explicit, consider the *lenient continuation game*, the subgame following the prosecutor's choice of $\sigma = \text{lenient}$. Fix a category i , and let $z \in \{0, 1\}$ denote the prosecutor's discretionary treatment of low-signal cases, with $z = 1$ if $x(i, L) = 1$, and let $e \in [0, \bar{e}]$ denote police evidence investment. Let $\Phi_L(e)$ denote $\Phi(i, L; \text{lenient})$ evaluated at the distribution of case signals caused by e . Given e , the prosecutor charges a low-signal case

whenever $\Phi_L(e) > 0$ and declines it when $\Phi_L(e) < 0$. Given z , the police choose e to maximize their expected return net of investigative costs.

The possibility of multiplicity comes from the interaction between these choices. When the prosecutor charges low-signal cases, evidence investment does not affect whether a case is charged. The value to police of investment therefore comes only from its consequences for conviction, producing a relatively low investment level. If $\Phi_L(e)$ is positive at that level, broad charging is optimal and so the low-investment outcome can be sustained in equilibrium. When the prosecutor instead declines low-signal cases, investment is valuable to the police not just because of its effect on conviction but also because it raises the probability that a case produces a high signal and is charged. That makes investment more attractive, comparatively, and so police may choose a higher investment level. If $\Phi_L(e)$ is negative at that level, selective charging is optimal and so the high investment outcome can be supported in equilibrium. Thus, the same lenient standard can support both a low-investment, broad-charging equilibrium and a high-investment, selective-charging equilibrium.

Proposition 4 (Multiplicity under lenient screening) *Consider category i and consider the police's effort choice in isolation under $\sigma = \text{lenient}$. Let $z \in \{0, 1\}$ denote the prosecutor's discretionary choice to charge low-signal cases, $z = 1 \iff x(i, L) = 1$, and let $e \in [0, \bar{e}]$ denote police evidence investment in that category. Define the prosecutor's low-signal net gain $\Phi_L(e) \equiv \Phi(i, L; \text{lenient})$, where the posterior $\mu_i(L; \text{lenient})$ is computed at the signal distribution induced by e . Let $BR_P(e)$ be the prosecutor's best-response correspondence for z , and let $BR_L(z)$ be the police best-response correspondence for e .*

Assume:

- (i) (Monotone incentives) $\Phi_L(e)$ is strictly decreasing in e .
- (ii) (Investment complementarity) For any $e \in BR_L(0)$ and any $e' \in BR_L(1)$, we have $e \geq e'$, with strict inequality for interior solutions.
- (iii) (Crossing) $\Phi_L(\min BR_L(1)) > 0$ and $\Phi_L(\max BR_L(0)) < 0$.

Then the reduced one-category problem admits at least two PBE-consistent fixed points (z, e) :

1. a low-investment, broad-charging fixed point with $z = 1$ and $e \in BR_L(1)$; and
2. a high-investment, selective-charging fixed point with $z = 0$ and $e \in BR_L(0)$.

Each fixed point can be part of a PBE with a lenient standard if police and charging decisions

in both categories satisfy the conditions in Appendix A-2.1 and the prosecutor weakly prefers leniency to strictness. Both comparisons use the same equilibrium behavior under a strict standard.

Proposition 4 isolates the coordination logic that drives the results I describe in the next section. In particular, it implies that identical formal standards within the lenient regime can be associated with different effective screening and evidentiary environments. Thus, across a wide range of substantive settings, a lenient formal prosecutorial standard can involve either selective screening and its associated pattern of police investment and resource allocation or an unscreened regime and its associated police effort and resource allocation.

4 Coordination and Institutional Path Dependence

I now turn from characterizing the equilibria to describing their implications for reform. Central to this analysis is the potential for multiplicity, as it focuses our attention on how history shapes criminal justice reform and how we can interpret empirical analyses of the legal system.

4.1 Comparative statics and equilibrium behavior

To begin, consider how various model parameters determine what kind of equilibrium behavior can be supported. The regimes identified above arise from the interaction of three forces: electoral incentives, the costs of weak prosecutions, and the productivity of police evidence investment. These same forces also determine whether there is a unique or multiple equilibria.

Given a lenient standard, charging a weak files in category i requires $\Phi(i, L; \text{lenient}) \geq 0$. This applies to both categories in the unscreened equilibrium and to category A in Proposition 3. Let

$$\mu_i^L = \frac{\pi_i(1 - \alpha_i(e_i^L))}{\pi_i(1 - \alpha_i(e_i^L)) + (1 - \pi_i)(1 - \beta_i(e_i^L))}$$

denote the posterior probability of guilt after a weak file at the specified continuation. Its

conviction probability is $Q_i^L = \mu_i^L q(1, L) + (1 - \mu_i^L)q(0, L)$. We can then express the weak-file charging condition as:

$$(b + a)Q_i^L + \Delta_E(i, L; \text{lenient}) \geq a + \lambda(1 - \mu_i^L)q(0, L). \quad (5)$$

Thus, increasing b or the increase in incremental electoral return makes charging more attractive, while increasing a or λ makes it less attractive. We cannot rank the prosecutor's ex ante preference, though, with these conditional effects alone. Doing so depends on the different portfolios induced under each standard. (Proposition 7 in Appendix A-3.2 gives the derivatives and distinguishes conditional charging incentives from policy comparisons.)

4.2 Strategic complementarity and institutional path dependence

Proposition 4's conditions define a region in which a low-investment, broad-charging equilibrium and a high-investment, selective-charging equilibrium coexist under the same lenient standard.

The mechanism follows from Condition (5). Under the conditions defined in Proposition 7, higher police investment makes a low signal more diagnostic of innocence and reduces the prosecutor's return to charging the case. When investment is low, broad charging is optimal. Broad charging in turn removes the charging-margin return to evidence, inducing police to maintain low investment. When investment is high, selective charging is optimal. The prosecutor's screening of cases increases the incentive for the police to invest in evidence, because investment raises the probability that a case produces a chargeable high signal. Because it is the charging rules that shape the equilibrium, inherited expectations can sustain coordination that persists after a reform if that reform leaves the previous fixed point still viable. It is in this sense that path dependence comes about from equilibrium multiplicity and allows reform to produce no observable change in outcomes or behavior.

4.3 Screening, reallocation, and reform

This coordination logic interacts with the cross-category priority mechanism emphasized above. Under strict screening, evidence investment affects not only conviction probability but also the probability of being charged, which increases the marginal return to investigative effort in categories where evidence is productive. The result can be a reallocation in which changing from a lenient to a strict standard induces the police to shift enforcement toward categories whose relative net returns increase.⁴ This reallocation of cases across areas of enforcement means that changing charging standards can directly shape the set of cases that police produce for prosecution in the first place. Empirically this means that even if a reform tightens formal standards, police behavior and observed outcomes may remain similar if the reform does not shift which of the two fixed points the system coordinates on.

Those observations in turn have direct bearings on empirical analyses of prosecutorial reform and its effects on the criminal justice system. For example, Agan et al. (2025) find that the election of reform prosecutors leads to reduced charging and conviction rates in misdemeanor cases without a similar effect in felony cases. Mitchell et al. (2022) document evidence that selecting a reform prosecutor leads to reduced racial disparities in who is charged with a crime. Further, this finding has immediate implications for the nascent literature that examines how police priorities and enforcement efforts respond to expectations about how suspected criminals will be treated in later stages of the criminal justice system (e.g., Goncalves and Mello Forthcoming, Soliman 2022). In all of these instances, our inferences from empirical relationships between policy reform and policing and prosecutorial decisions are complicated by the coordination dynamic that drives equilibrium selection.

4.4 Reform with and without change

First consider reform without observed change. In this situation, we have persistence of behavioral patterns that is due to multiplicity of equilibria that remains after reform. To state the

⁴I give sufficient conditions for this in Proposition 9 in Appendix A-3.

formal result cleanly, define reform as a parameter shift that affects payoffs or institutions but not necessarily equilibrium selection.

Definition 1 (Reform as a parameter shift) *Let ω denote the relevant primitives (e.g., $a, \lambda, \rho, \Gamma, \{\alpha_i, \beta_i\}$, the effort budget $\bar{B}$, and the reallocation cost C). A reform is a shift $\omega \rightarrow \omega'$. Let $\mathcal{E}(\omega)$ denote the set of PBE under primitives ω .*

Under this definition, reform might be thought of as any change in the players' preferences or the technology that maps police effort into case files.

Proposition 5 (Persistence after reform)

Fix a category i and suppose that under ω there are two PBEs that involve a lenient standard, with respective behavior in category i : a low-investment/broad-charging equilibrium $(z, e) = (1, e_L)$ and a high-investment/selective-charging equilibrium $(z, e) = (0, e_H)$ with $e_H > e_L$. In each PBE, assume strict weak-file incentives and a strict initial preference for leniency.

Consider a sufficiently small reform $\omega \rightarrow \omega'$ such that:

- (i) only prosecutor payoff parameters change, with conditional electoral increments and policy values continuous in those parameters, and*
- (ii) post-reform behavior remains coordinated on the pre-reform equilibrium (equilibrium persistence).*

Then both PBEs persist, and the reform leaves observed aggregate outcomes (e.g., charging rates, conviction rates, declination rates) unchanged, even though primitives have shifted.

Proposition 5 reveals the way in which the model predicts institutional persistence following a change in prosecutorial incentives. Rather than a more standard story about bureaucratic inertia, this result shows that the same fixed point can be supported under different prosecutorial incentives. Marginal increases in a prosecutor's aversion to wrongful convictions, for example, may still be able to sustain a low investment equilibrium and, if so, then police will have no incentive to shift their behavior simply because the prosecutor has changed. Thus, selecting a more "progressive" prosecutor, who places greater weight on avoiding wrongful convictions, does not necessarily lead to a change in equilibrium policing and charging.

Next, consider when reform *must* bring about behavioral change. That turns on the prosecu-

tor's incentives to charge a case change. In particular, increasing the cost of failed prosecutions can induce the prosecutor to not charge a weak case, while leaving charging decisions unchanged in an office that already declines them. The consequences of reform therefore depend in part on the enforcement practices already in place.

Proposition 6 (Behavioral change after reform)

Fix i and suppose multiplicity obtains under ω as in Proposition 5. If a reform $\omega \rightarrow \omega'$ shifts primitives so that, in the post-reform lenient continuation game, an equilibrium exists and either

$$\begin{aligned} \Phi_i(\xi, z; \omega') &> 0 \quad \text{for every } z \text{ with } z_i = 0 \text{ and every interior } \xi \in \mathcal{B}(z; \omega'), \\ &\text{or} \\ \Phi_i(\xi, z; \omega') &< 0 \quad \text{for every } z \text{ with } z_i = 1 \text{ and every interior } \xi \in \mathcal{B}(z; \omega'), \end{aligned}$$

then the lenient continuation game under ω' rules out declining or charging weak files in category i , respectively, and equilibrium behavior must change relative to at least one pre-reform equilibrium. Here $z_i = x(i, L)$ and $\mathcal{B}(z; \omega')$ contains the jointly optimal police choices given charging rule z , as defined in Appendix A-2.1.

Proposition 6 reveals a key distinction in types of criminal justice reforms. Some reforms simply *tilt* incentives, whereas others *change the structure of the coordination problem*. Changes in the net expected value from charging a weak case ($\Phi_L(e)$) can be brought about by increasing courtroom scrutiny (sensitivity of q to the state, θ), increasing the electoral penalty for failed prosecutions through Γ , raising the costs a or λ , or improving the diagnostic productivity of evidence production. However, due to equilibrium persistence, those changes have to be large enough to actually change the equilibrium in order to change observed behavior.

4.5 Reinterpreting existing empirical findings

These theoretical results have direct implications for how we should interpret the kinds of empirical analyses of policing and prosecution that are common in the extant literature. Typical studies investigate how reform affects outcomes such as charging rates and conviction rates. However, as we have seen here, these outcomes are themselves equilibrium phenomena shaped by both prosecutorial screening and police behavior. Thus, we cannot simply attribute variation in those kinds of quantities to prosecutorial policy alone. At a minimum, these outcomes reflect

both the selection of cases into prosecution and the distribution of case quality induced by police priorities and evidence investment.

Selection and conditional conviction. First, there is a direct prediction concerning how screening selects cases for prosecution. The model predicts that, within a category and holding police effort fixed, a reform that moves towards stricter evidentiary screening reduces charging rates without decreasing the conviction rate conditional on charge. This is because stricter screening draws charged cases from a stronger pool. That selection effect can arise even if case quality is fixed or invariant to the screening standard. This is a prediction that can easily be assessed with the kinds of data that are commonly used in analyses of prosecution and reform (e.g., Agan et al. 2025, Mitchell et al. 2022). To look for evidence consistent with an effect of the screening standard on the mix of cases, we would consider whether the conditional conviction rate changes with a reform.

Heterogeneity across offense categories. The model’s second implication for empirical analysis has to do with how reform’s effects vary across types of offenses, a topic of considerable empirical interest (e.g., Agan, Doleac and Harvey 2023, Agan et al. 2025, Mitchell et al. 2022). In the model, variation in police effort and prioritization across areas of law enforcement arises endogenously (i.e., the category-dependent equilibrium). This is because different kinds of offenses might vary in baseline convictability, courtroom scrutiny, or the productivity of evidence investment. As a result, screening can differ across areas, giving police different incentives to investigate, and therefore the same reform may have different effects across offense types. In brief, the model’s equilibria explicitly predict the kind of heterogeneity that the extant literature examines.

Divergent reform outcomes. It might be easy or natural to interpret the contrast between Philadelphia and San Francisco, which I described at the start of this paper, as a story of one reform succeeding and another failing. However, the model I develop here offers an alternative interpretation. In the model similar formal reforms can yield divergent system-level

consequences simply because jurisdictions may have started in different equilibria of the coordination game. That interpretation has a testable implication that distinguishes it from the success-or-failure account. Specifically, the model predicts that pre-reform enforcement patterns, and in particular the pre-reform sensitivity of charging to evidentiary strength, should predict the magnitude of the post-reform response. Indeed, some existing work that examines examining whether police responded endogenously to Philadelphia’s reforms (Ouss and Stevenson 2023) is consistent with that mechanism. That line of work, though, is not designed to be able to test the prediction, which is a valuable direction for future empirical work on the politics of criminal justice reform.

How police respond. To the extent recent empirical work has attended to the endogeneity of police behavior to prosecutorial policy, it typically treats that endogeneity as a threat to identification and devotes analyses to detecting or ruling it out (Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023). From the perspective of the model I develop here, police responsiveness is itself a structural feature rather than a confound, with theoretical predictions about the nature of that response. The model predicts that police may reallocate enforcement across offense categories as the relative net returns to pursuing cases change, and we may observe changes in how they invest in cases within categories related to how the screening standard changes (e.g., Proposition 9). That insight bridges the nascent literature on how police effort responds to expected downstream treatment (e.g., Goncalves and Mello Forthcoming, Soliman 2022) with the literature on prosecutorial reform. What is more, it brings specific predictions to what are typically treated as open-ended robustness questions about the endogeneity of policing.

Across these various empirical reinterpretations, there is a common structure. The various equilibria I study differ not only in average outcomes but also in the structure of enforcement. They differ in how charging decisions respond to a case’s evidentiary strength, how police effort is allocated across categories, and how informative prosecutorial decisions are about underlying legal merit. These are the sources of variation across equilibria that allow us to empirically

discriminate among different kinds of equilibria.

Implication 1 (Observable regime signatures) *Across the equilibrium regimes, the model predicts:*

1. *The unscreened regime produces high charging rates and no sensitivity of charging to evidentiary strength.*
2. *The screened regime produces lower charging rates and strong sensitivity to evidence.*
3. *The category-dependent regime produces heterogeneous charging thresholds across categories.*

Implication 1 identifies how we can empirically discriminate among equilibria and also point towards how best to structure measurement for an analysis of reform. The most informative quantities are the sensitivity of charging to evidentiary strength and the allocation of enforcement across categories. These are things that the contemporary empirical literature lends itself to measuring using, for example, information about body-camera-compliance and corroboration indicators, case-file completeness, recorded dismissal reasons, and the composition of arrests across offense types. And, if that is feasible, then we may very well be feasible to estimate how charging decisions respond to case strength by measuring these kinds of quantities along with charging and conviction outcome.

However, Proposition 5 goes further in that it helps interpret various potential null findings. A meaningful reform may nevertheless leave aggregate charging and conviction rates unchanged when the system remains coordinated on a pre-reform equilibrium. That persistence, then, predicts a null effect, rather than casting it as evidence that prosecutorial discretion is inconsequential. What Implication 1 does is help us measure whether the structure of enforcement persists. Under persistence, the structure of enforcement is preserved along with the aggregates.

Of course, there are limits to what these results allow us to infer from data about policing and prosecution. First, the discussion here is about interpretation rather than identification. Simply because we observe patterns consistent with the model's mechanisms does not rule out other possible explanations. Other theories that focus on criminal justices resources or institutional preferences, for example, may yield predictions observationally equivalent with those I outline here. Second, the model's sharpest predictions concern the sensitivity of charging to case

strength. But, to the extent that we cannot measure that sensitivity, the model only makes predictions about the direction of the relationship, rather than its magnitude. That having been said, the model’s key empirical implication is that it predicts pre-reform enforcement patterns should predict post-reform responses. This is the benefit of conceiving of policing and prosecution as strategic complements.

5 Discussion and Conclusion

Taken together, the analysis here helps recast our understanding of the politics of criminal justice by highlighting the implications of strategic coordination dynamic that arises as a result of the complementary functions that police and prosecutors play.

Strategic interdependence and institutional governance. The most immediate implication from the analysis is that prosecution and policing are jointly determined equilibrium outcomes rather than sequential stages of enforcement. As I have argued, we should understand prosecutorial standards and police evidence production as strategic complements. That means that l authority over charging does not translate mechanically into control over enforcement outcomes.

Three implications of that interdependence bear discussion. First, authority should be decoupled from influence in the way we understand the politics of law enforcement. Prosecutors set charging rules, but police determine the informational environment in which those rules operate. Police confront a particular set of incentives that is shaped by the screening rules prosecutors apply to the cases the police bring to them. In turn, those incentives shape what the prosecutors should infer about police effort and the cases they have to handle. In this sense, policy is jointly produced in equilibrium rather than imposed unilaterally. Where past research emphasizes police influence over their own policies, their lobbying, and their enforcement priorities (e.g., Cheng 2024, Eckhouse 2022, Goncalves and Mello Forthcoming, Mummolo 2018), this model reveals a subtler, systemic channel in which neither actor shapes outcomes unilaterally.

Second, the strategic complementarity between policing and prosecution suggests equilibrium patterns, rather than a simple marginal change in either. Most pointedly, the model reveals that identical formal standards can sustain qualitatively distinct enforcement environments depending on how police and prosecutors coordinate. That has broad implications for policy evaluation and studies of reform effects on system performance and individual welfare (e.g., Agan et al. 2025, Bandyopadhyay and McCannon 2014, Dobbie, Goldin and Yang 2018).

Third, the analysis shows how strategic interdependence can lead to institutional path dependence. This boils down to the possibility that a reform may shift payoffs and incentives but nevertheless leave the coordination problem intact, which implies there may be little or no observable change in behavior. In that sense, stability in enforcement patterns can reflect equilibrium persistence rather than bureaucratic inertia (cf. Sklansky 2018). In short, reforms might fail without failing. More broadly, the model I present here reframes prosecutorial reform as a problem of institutional governance in which delegation operates through endogenous information production and control depends on how oversight rules reshape agent incentives upstream. These dynamics are likely to arise in other policy domains where formally accountable principals rely on semi-autonomous agents to generate the information on which policy decisions depend.

Prosecutorial reform is fundamentally a coordination problem. Police resistance to prosecutorial reform is often attributed to differences in preferences about how to treat suspected criminals or tastes for punitiveness. The model highlights a subtler reason. If a prosecutor demands stronger evidence, police are required to engage in costly effort to build stronger files, and even selectively strict standards endogenously shift both the amount of police effort and its allocation across areas of law enforcement.

Future theoretical and empirical research. Several extensions are beyond this paper’s scope. Because police are politically active actors (e.g., Cheng 2024), as the Krasner and Boudin recall and election fights illustrate, a natural next step is to let players anticipate who

becomes prosecutor if the incumbent loses. Another is to let potential offenders anticipate the coordination dynamics, endogenizing the mix of cases available for arrest and prosecution.

Finally, the analysis points to a clear agenda for empirical work, developed above. Future research should study how the sensitivity of charging to evidentiary strength responds to reform rather than average treatment effects alone, treat policing as endogenous to prosecutorial policy, and read null or modest reform effects as possible equilibrium persistence rather than the irrelevance of prosecutorial discretion.

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Coordinated Justice:
The Politics of Prosecution and Policing

Online Appendix

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Appendix

A-1 Model objects and parameters

Object	Definition
$i \in \{A, B\}$	Case category; its probability is $p_A(r) = r$ or $p_B(r) = 1 - r$.
σ	Announced evidentiary standard, strict or lenient.
$\mathcal{X}_\sigma$	Admissible charging rules: strong files are charged under either standard; weak files must be declined under strictness.
$z_i = x(i, L)$	Whether weak files in category i are charged; $z = (z_A, z_B)$.
$r \in [0, 1]$	The commissioner's allocation of cases to category A .
$e_i \in [0, \bar{e}]$	Investigative effort per case in category i .
$\bar{B}$	Capacity: $re_A + (1 - r)e_B \leq \bar{B}$.
$\theta \in \{0, 1\}, \pi_i$	Defendant's guilt and the prior $\Pr(\theta = 1 \mid i)$.
$s \in \{H, L\}$	Strong or weak evidence observed by the prosecutor.
$\alpha_i(e), \beta_i(e)$	Probabilities of strong evidence conditional on guilt or innocence.
$x, y \in \{0, 1\}$	Charging and conviction; set $y = 0$ when $x = 0$.
$q(\theta, s)$	Conviction probability conditional on charging, guilt, and evidence.
b, a, λ	Prosecutorial benefit of conviction, cost of acquittal, and cost of wrongful conviction.
ρ, Γ	Value of retention and the electoral evaluation rule.
$v_i > 0, w_i \geq 0$	Police values of a charge and a conviction.
$c(e), C(r)$	Per-case investigation cost and the cost of reallocating enforcement.
$\mu_i(s), \mu$	Posterior probability of guilt and the shadow price of police capacity, respectively.
$\Delta_E(i, s; \sigma)$	Expected electoral payoff from charging minus that from declining.
Φ_i	Net payoff from charging a weak file, evaluated at the specified continuation.
$\mathcal{B}(z)$	Jointly optimal police allocation and effort given charging profile z .
$V_P(\sigma; k)$	Prosecutor's payoff under standard σ and specified continuation k .

Table A-1: Model objects and parameters. Effort is per case; total effort in a category is $p_i(r)e_i$.

A-2 Proofs of the results in the main text

The proofs follow the order of the main-text results. I first define the probability and payoff expressions used below. Auxiliary results and comparative statics follow the main-text proofs.

A-2.1 Notation and the equilibrium assessment

Recall that by assumption $\bar{e} > 0$, $\bar{B} > 0$, $0 < \pi_i < 1$, and the signal-generation technologies are continuously differentiable, satisfying $0 < \beta_i(e) < \alpha_i(e) < 1$ on $[0, \bar{e}]$. Costs are continuous, with the differentiability stated wherever first-order conditions are used. The equilibrium proofs concern interior case allocations, $0 < r < 1$. This is a restriction on the candidate police solutions, not a consequence of convex reallocation costs. It ensures that both categories and both signals are reached. Boundary allocations are discussed in Appendix A-3.1.

Probabilities and payoffs. Write $z = (z_A, z_B)$ for the weak-file charging profile, where $z_i = x(i, L)$ equals one if the prosecutor charges a weak file in category i and zero if she declines it. Strong files are charged in both categories. Let $p_A(r) = r$ and $p_B(r) = 1 - r$ be the category probabilities, and write $e = (e_A, e_B)$ for the pair of per-case effort choices. Finally, $q_{\theta s} = q(\theta, s)$ is the conviction probability conditional on charging, guilt state θ , and signal s . Define

$$h_i(e) = \pi_i \alpha_i(e) + (1 - \pi_i) \beta_i(e), \quad (\text{A-1})$$

$$t_{iH}(e) = \pi_i \alpha_i(e) q_{1H} + (1 - \pi_i) \beta_i(e) q_{0H}, \quad (\text{A-2})$$

$$t_{iL}(e) = \pi_i [1 - \alpha_i(e)] q_{1L} + (1 - \pi_i) [1 - \beta_i(e)] q_{0L}. \quad (\text{A-3})$$

Here $h_i(e)$ is the probability of a strong file, and $t_{is}(e)$ is the joint probability of signal s and a conviction if files with that signal are charged. All three are conditional on category i and effort e . For a weak-file charging decision $z_i \in \{0, 1\}$, the charging rate, conviction rate, and wrongful-conviction rate within category i are

$$P_i^{z_i}(e) = h_i(e) + z_i [1 - h_i(e)], \quad (\text{A-4})$$

$$Y_i^{z_i}(e) = t_{iH}(e) + z_i t_{iL}(e), \quad (\text{A-5})$$

$$W_i^{z_i}(e) = (1 - \pi_i) \{ \beta_i(e) q_{0H} + z_i [1 - \beta_i(e)] q_{0L} \}. \quad (\text{A-6})$$

Failed prosecutions have probability $P_i^{z_i} - Y_i^{z_i}$; declinations have probability $1 - P_i^{z_i}$. The police net return per case is

$$g_i^{z_i}(e) = v_i P_i^{z_i}(e) + w_i Y_i^{z_i}(e) - c(e). \quad (\text{A-7})$$

For an allocation r , define the feasible effort set and officer response by

$$\mathcal{F}(r) = \{(e_A, e_B) \in [0, \bar{e}]^2 : r e_A + (1 - r) e_B \leq \bar{B}\}, \quad (\text{A-8})$$

$$E_z(r) = \operatorname{argmax}_{e \in \mathcal{F}(r)} \{r g_A^{z_A}(e_A) + (1 - r) g_B^{z_B}(e_B)\}. \quad (\text{A-9})$$

The commissioner chooses a global maximizer of

$$F_z(r) = \max_{e \in \mathcal{F}(r)} \{r g_A^{z_A}(e_A) + (1-r) g_B^{z_B}(e_B)\} - C(r). \quad (\text{A-10})$$

Let $\mathcal{B}(z)$ contain the resulting triples $\xi = (r, e_A, e_B)$, with $e \in E_z(r)$. This definition optimizes effort at every candidate allocation; it does not hold effort fixed when the commissioner deviates. Continuity and compactness give nonempty maximizing sets. Appendix A-3.1 establishes this claim and gives sufficient conditions for uniqueness. The constructions use global maximizers even when first-order conditions are insufficient.

Beliefs and charging. At an interior candidate ξ , Bayes' rule gives the prosecutor's posterior probability that the defendant is guilty

$$\mu_i(H; \xi) = \frac{\pi_i \alpha_i(e_i)}{h_i(e_i)}, \quad (\text{A-11})$$

$$\mu_i(L; \xi) = \frac{\pi_i [1 - \alpha_i(e_i)]}{1 - h_i(e_i)}. \quad (\text{A-12})$$

Let $Q_i(s; \xi) = \mu_i(s; \xi) q_{1s} + [1 - \mu_i(s; \xi)] q_{0s}$ be the probability of conviction if the prosecutor charges the case. The argument $\xi = (r, e_A, e_B)$ identifies the police behavior anticipated in equilibrium; the prosecutor observes the category and signal, but not ξ . Write $\Delta_E(i, s; \sigma)$ for the expected electoral payoff from charging minus that from declining, as defined in the main text. The weak-file charging gain is

$$\Phi_i(\xi, z; \sigma) = (b + a) Q_i(L; \xi) - a - \lambda [1 - \mu_i(L; \xi)] q_{0L} + \Delta_E(i, L; \sigma). \quad (\text{A-13})$$

The arguments (ξ, z) make explicit the dependence on police choices and the charging profile suppressed in $\Phi(i, L; \sigma)$.

Under leniency, a candidate is sequentially rational at the charging stage exactly when, for each i ,

$$z_i = 1 \implies \Phi_i(\xi, z; \text{lenient}) \geq 0, \quad z_i = 0 \implies \Phi_i(\xi, z; \text{lenient}) \leq 0. \quad (\text{A-14})$$

Both decisions are optimal at equality. Under strictness, weak-file charging is infeasible. Strong-file charging is prescribed by either admissible set; the maintained positive strong-file payoff is consistent with that prescription but is not needed to rule out an infeasible deviation.

The electoral increment in equation (A-13) is the actual conditional payoff difference in the representative-case game. When elections are interpreted through a portfolio, expectations integrate over the remaining cases in that portfolio. The same electoral specification must be used to compute both Δ_E and the ex ante payoff below. The proof treats charging as the case-level decision in the stated timing. If the prosecutor instead chose a whole portfolio rule at once under a nonlinear Γ , separate case-level sign tests would not in general suffice for that different optimization problem.

Strategies after either policy. A continuation k_σ specifies a charging profile z^σ , an officer selection $e^\sigma(r) \in E_{z^\sigma}(r)$ at every r , and a maximizing allocation r^σ . Select the officer response

at r^σ to coincide with the candidate in $\mathcal{B}(z^\sigma)$. Beliefs are given by equations (A-11)–(A-12). The prosecutor observes σ and (i, s) , but not the police choices. A hidden police deviation changes the distribution of signals; it does not change her stipulated charging strategy or make her observe the deviation.

For a continuation k , the aggregate charging, conviction, and wrongful-conviction rates are $\bar{P} = \sum_i p_i(r) P_i^{z_i}(e_i)$, $\bar{Y} = \sum_i p_i(r) Y_i^{z_i}(e_i)$, and $\bar{W} = \sum_i p_i(r) W_i^{z_i}(e_i)$. Let $R_\sigma(k) = \mathbb{E}[\Gamma(y, x(1 - y), 1 - x, \sigma) \mid k]$ be the expected retention probability. The prosecutor's expected payoff before choosing the standard is

$$V_P(\sigma; k) = b\bar{Y} - a(\bar{P} - \bar{Y}) - \lambda\bar{W} + \rho R_\sigma(k). \quad (\text{A-15})$$

The initial policy maximizes this expression over the two *specified* continuations. In particular, a deviation to another standard invokes the continuation prescribed there. It does not allow the prosecutor to select whichever subsequent equilibrium gives her the highest payoff. A full PBE therefore includes a valid continuation after the policy that is not chosen.

A-2.2 Proofs

Proof of Lemma 1. I prove necessity using the first-order conditions and sufficiency using concavity. Fix the prosecutor's charging rule and let $z_i = x(i, L) \in \{0, 1\}$ denote weak-file charging in category i . Write

$$g_i(e) = v_i\{h_i(e) + z_i[1 - h_i(e)]\} + w_i[t_{iH}(e) + z_i t_{iL}(e)] - c(e)$$

for the police return per case net of investigation costs. Here $h_i(e)$ is the probability of $s = H$, and $t_{is}(e)$ is the joint probability of signal s and $y = 1$ when $x(i, s) = 1$, as defined in equations (A-1)–(A-3).

For any r , officers maximize $rg_A(e_A) + (1 - r)g_B(e_B)$ subject to the effort bounds and capacity constraint. Let $\mu \geq 0$ be the multiplier on $re_A + (1 - r)e_B \leq \bar{B}$. Since $0 < r < 1$, both category weights are positive. Dividing each effort condition by its weight gives $g'_i(e_i) = \mu$ for $0 < e_i < \bar{e}$, $g'_i(0) \leq \mu$ at $e_i = 0$, and $g'_i(\bar{e}) \geq \mu$ at $e_i = \bar{e}$. Differentiating g_i gives equation (2) when $z_i = 1$ and equation (3) when $z_i = 0$. Complementary slackness gives $\mu[\bar{B} - re_A - (1 - r)e_B] = 0$. Because the commissioner and officers share an objective, their optimal choices also maximize the joint police payoff. Its allocation condition at $0 < r < 1$ is

$$g_A(e_A) - g_B(e_B) - \mu(e_A - e_B) = C'(r),$$

which is equation (1). This establishes necessity.

For sufficiency, take feasible police choices (r^*, e_A^*, e_B^*) satisfying these conditions, and define $\ell_i = g_i(e_i^*) - \mu e_i^*$, the net return per case after valuing effort at the shadow price μ . By the assumed concavity of g_i , the effort conditions imply $g_i(e) - \mu e \leq \ell_i$ for every $e \in [0, \bar{e}]$. Therefore any feasible alternative (r, e_A, e_B) yields a police payoff no greater than

$$r\ell_A + (1 - r)\ell_B + \mu\bar{B} - C(r).$$

The allocation condition gives $C'(r^*) = \ell_A - \ell_B$. Convexity of C makes r^* a global maximizer

of this upper bound. The choices (r^*, e_A^*, e_B^*) attain the bound by complementary slackness and the definition of ℓ_i , so they maximize the joint police payoff. Let the commissioner choose r^* and officers choose (e_A^*, e_B^*) there. For each $r \neq r^*$, let officers maximize their payoff subject to the effort bounds and capacity constraint. These strategies are sequentially optimal for the fixed charging rule. $\blacksquare$

Proof of Proposition 1. I prove existence by construction. For each $\sigma \in \{\text{lenient}, \text{strict}\}$, let $z^\sigma = (z_A^\sigma, z_B^\sigma)$ denote weak-file charging, with $z_i^\sigma = x(i, L)$. Prescribe $z^{\text{lenient}} = (1, 1)$ and $z^{\text{strict}} = (0, 0)$, with $x(i, H) = 1$ for each i under both standards. Let the police follow optimal strategies for these charging rules. Their choices satisfy the necessary conditions in Lemma 1. The proposition's payoff inequalities are evaluated at these strategies and the beliefs they induce.

Let $h_i(e)$ be the probability of $s = H$ and $t_{is}(e)$ the joint probability of signal s and $y = 1$ when $x(i, s) = 1$, from equations (A-1)–(A-3). Given $z_i \in \{0, 1\}$, the police returns per case net of costs are

$$g_i^1(e) = v_i + w_i[t_{iH}(e) + t_{iL}(e)] - c(e), \quad g_i^0(e) = v_i h_i(e) + w_i t_{iH}(e) - c(e).$$

The superscript records z_i . For every σ and r , let officers choose $e^\sigma(r) = (e_A^\sigma(r), e_B^\sigma(r)) \in E_{z^\sigma}(r)$, the set of efforts maximizing their payoff subject to the effort bounds and capacity constraint. Let the commissioner choose $r^\sigma \in \arg\max_{r \in [0, 1]} F_{z^\sigma}(r)$, where F_{z^σ} is the payoff after officers optimize, defined in equation (A-10). Lemma 2 establishes that these maximizers exist. As assumed in Appendix A-2.1, $0 < r^\sigma < 1$. Officers therefore act optimally at every r , and the commissioner cannot gain by deviating from r^σ and inducing a different effort response. In the proposed equilibrium with $\sigma = \text{lenient}$, $x(i, H) = x(i, L) = 1$ for each i , so effort affects the police payoff only through convictions and costs. For $0 < e_i < \bar{e}$, differentiating g_i^1 gives

$$w_i \{ \pi_i \alpha'_i(e_i) [q(1, H) - q(1, L)] + (1 - \pi_i) \beta'_i(e_i) [q(0, H) - q(0, L)] \} = c'(e_i) + \mu, \quad (\text{A-16})$$

where $\mu \geq 0$ is the common capacity multiplier. Lemma 1 supplies the effort-bound inequalities and allocation condition. The global maximizations above establish optimality even when first-order conditions alone are insufficient.

Let $\xi^\sigma = (r^\sigma, e_A^\sigma(r^\sigma), e_B^\sigma(r^\sigma))$ denote the police choices following σ . The prosecutor expects the police to choose ξ^σ . Conditional on (i, s) , Bayes' rule gives $\mu_i(s; \xi^\sigma)$, the posterior probability of guilt in equations (A-11)–(A-12). Since $0 < r^\sigma < 1$ and $0 < h_i(e_i) < 1$, every (i, s) occurs with positive probability after either σ . Condition (a), evaluated at ξ^{lenient} and these beliefs, gives $\Phi(i, L; \text{lenient}) \geq 0$, so $z_i^{\text{lenient}} = 1$ is optimal for each i . The admissible rules require $x(i, H) = 1$ under both standards and $z_i^{\text{strict}} = 0$. The prosecutor does not observe police deviations, so $x(i, s)$ remains fixed when police deviate from ξ^σ . The police maximizations therefore evaluate their actual deviation payoffs.

Let k_{lenient} and k_{strict} denote the continuation strategies and beliefs following $\sigma = \text{lenient}$ and $\sigma = \text{strict}$, respectively. Using equation (A-15), condition (b) is

$$V_P(\text{lenient}; k_{\text{lenient}}) \geq V_P(\text{strict}; k_{\text{strict}}),$$

where V_P is the prosecutor's expected payoff. Thus choosing $\sigma = \text{lenient}$ is optimal, including

at equality. Strategies are sequentially rational after either σ , and beliefs satisfy Bayes' rule. These strategies and beliefs form a PBE. $\blacksquare$

Proof of Proposition 2. I prove existence by construction. After $\sigma = \text{strict}$, prescribe $x(i, H) = 1$ and $x(i, L) = 0$ for each $i \in \{A, B\}$. Equivalently, $z = (z_A, z_B) = (0, 0)$, where $z_i = x(i, L)$.

Let $h_i(e)$ be the probability of $s = H$ and $t_{iH}(e)$ the joint probability of $s = H$ and $y = 1$ when $x(i, H) = 1$, from equations (A-1)–(A-2). Given $z = (0, 0)$, the police net return per case is $g_i^0(e) = v_i h_i(e) + w_i t_{iH}(e) - c(e)$. At every r , let officers choose $e^{\text{strict}}(r) \in E_{(0,0)}(r)$ to maximize their payoff subject to the effort bounds and capacity constraint. Let the commissioner choose $r^{\text{strict}} \in \arg\max_{r \in [0,1]} F_{(0,0)}(r)$, where $F_{(0,0)}$ is the resulting payoff in equation (A-10). Lemma 2 establishes that these maximizers exist. As assumed in Appendix A-2.1, $0 < r^{\text{strict}} < 1$. For $0 < e_i < \bar{e}$, differentiating g_i^0 gives

$$v_i[\pi_i \alpha'_i(e_i) + (1 - \pi_i) \beta'_i(e_i)] + w_i[\pi_i \alpha'_i(e_i) q(1, H) + (1 - \pi_i) \beta'_i(e_i) q(0, H)] = c'(e_i) + \mu, \quad (\text{A-17})$$

where $\mu \geq 0$ is the common capacity multiplier. The allocation equation and effort-bound inequalities follow from Lemma 1. Because effort is optimized after every allocation, the commissioner cannot gain by changing priorities and inducing a different effort response.

Let $\xi^{\text{strict}} = (r^{\text{strict}}, e_A^{\text{strict}}(r^{\text{strict}}), e_B^{\text{strict}}(r^{\text{strict}}))$ denote these police choices. The prosecutor expects the police to choose ξ^{strict} . Conditional on (i, s) , Bayes' rule gives the posterior probability of guilt $\mu_i(s; \xi^{\text{strict}})$ in equations (A-11)–(A-12). Every (i, s) occurs with positive probability. Since $\sigma = \text{strict}$ requires $x(i, H) = 1$ and $x(i, L) = 0$, police behavior and charging are sequentially rational after this σ .

Following a deviation to $\sigma = \text{lenient}$, prescribe the continuation equilibrium k_{lenient} used to evaluate $V_P(\text{lenient})$ in the hypothesis. Let z^{lenient} denote its weak-file charging profile. At each r , let officers use the effort pair specified by k_{lenient} , which belongs to $E_{z^{\text{lenient}}}(r)$, and let the commissioner choose the allocation specified there, which maximizes $F_{z^{\text{lenient}}}(r)$. The prosecutor expects these police strategies and, conditional on (i, s) , updates the probability of guilt by Bayes' rule using equations (A-11)–(A-12). At these beliefs, her gain from $x(i, L) = 1$ rather than $x(i, L) = 0$ satisfies $\Phi(i, L; \text{lenient}) \geq 0$ if $z_i^{\text{lenient}} = 1$ and $\Phi(i, L; \text{lenient}) \leq 0$ if $z_i^{\text{lenient}} = 0$. Thus these strategies satisfy the continuation equilibrium conditions in Appendix A-2.1, although $\sigma = \text{lenient}$ is off path.

Let k_{strict} denote the continuation strategies and beliefs derived above for $\sigma = \text{strict}$. The hypothesis states that the prosecutor's expected payoff satisfies

$$V_P(\text{strict}; k_{\text{strict}}) > V_P(\text{lenient}; k_{\text{lenient}}).$$

She therefore chooses $\sigma = \text{strict}$. Strategies are sequentially rational after either σ , and beliefs satisfy Bayes' rule, establishing the claimed PBE. $\blacksquare$

Proof of Proposition 3. I prove existence by construction. Let $z = (z_A, z_B)$ denote weak-file charging, with $z_i = x(i, L)$. After $\sigma = \text{lenient}$, prescribe $z = (1, 0)$ and $x(i, H) = 1$ for each $i \in \{A, B\}$.

Let $h_i(e)$ be the probability of $s = H$ and $t_{is}(e)$ the joint probability of signal s and $y = 1$ when

$x(i, s) = 1$, from equations (A-1)–(A-3). Given $z = (1, 0)$, the net returns per case are

$$g_A^1(e) = v_A + w_A[t_{AH}(e) + t_{AL}(e)] - c(e), \quad g_B^0(e) = v_B h_B(e) + w_B t_{BH}(e) - c(e).$$

At every r , let officers choose $e^{\text{lenient}}(r) \in E_{(1,0)}(r)$ to maximize their payoff subject to the effort bounds and shared budget. Let the commissioner choose $r^{\text{lenient}} \in \arg\max_{r \in [0,1]} F_{(1,0)}(r)$, where $F_{(1,0)}$ is the resulting payoff in equation (A-10). As assumed in Appendix A-2.1, $0 < r^{\text{lenient}} < 1$. These optimal police choices satisfy the necessary conditions in Lemma 1: effort satisfies equation (2) in A and equation (3) in B when $0 < e_i < \bar{e}$, with the corresponding inequalities at the bounds and a common capacity multiplier $\mu \geq 0$. The commissioner's allocation condition is

$$C'(r) = g_A^1(e_A) - g_B^0(e_B) - \mu(e_A - e_B), \quad (\text{A-18})$$

evaluated at the chosen allocation and efforts. The global maximizations establish optimality at both police stages; the first-order equations describe the resulting choices.

Let $\xi^{\text{lenient}} = (r^{\text{lenient}}, e_A^{\text{lenient}}(r^{\text{lenient}}), e_B^{\text{lenient}}(r^{\text{lenient}}))$. The prosecutor expects the police to choose ξ^{lenient} . Conditional on (i, s) , Bayes' rule gives her posterior probability of guilt $\mu_i(s; \xi^{\text{lenient}})$ in equations (A-11)–(A-12). Let $\Phi(i, L; \text{lenient})$ be her payoff from $x(i, L) = 1$ minus that from $x(i, L) = 0$. The proposition's inequalities are evaluated at these police strategies and beliefs, so $\Phi(A, L; \text{lenient}) \geq 0$ and $\Phi(B, L; \text{lenient}) < 0$. Thus $z_A = 1$ and $z_B = 0$ are optimal, and $x(i, H) = 1$ is required. Since the prosecutor does not observe police choices, $x(i, s)$ remains fixed when police deviate. The maximizations above therefore show that neither officers nor the commissioner can gain by deviating.

Following a deviation to $\sigma = \text{strict}$, prescribe $z = (0, 0)$ and $x(i, H) = 1$. The police return becomes $g_i^0(e) = v_i h_i(e) + w_i t_{iH}(e) - c(e)$. At every r , let officers choose an effort pair from $E_{(0,0)}(r)$ that maximizes their payoff, and let the commissioner maximize $F_{(0,0)}(r)$ anticipating these responses. The prosecutor expects these police strategies and, conditional on (i, s) , updates the probability of guilt by Bayes' rule using equations (A-11)–(A-12). Since $\sigma = \text{strict}$ requires $x(i, H) = 1$ and $x(i, L) = 0$, these strategies are sequentially rational.

Let k_{lenient} and k_{strict} denote the continuation strategies and beliefs after $\sigma = \text{lenient}$ and $\sigma = \text{strict}$, respectively. The assumed payoff comparison, evaluated by equation (A-15), is $V_P(\text{lenient}; k_{\text{lenient}}) \geq V_P(\text{strict}; k_{\text{strict}})$. Thus the prosecutor optimally chooses $\sigma = \text{lenient}$. Optimal behavior after either σ and Bayesian beliefs establish the PBE. The choices $z_A = 1$ and $z_B = 0$ give different charging incentives for investigation; a ranking of e_A and e_B would require additional restrictions on returns and technologies. ■

Proof of Proposition 4. I construct two mutual best responses in the one-category problem, then verify the conditions for extending them to PBEs in the two-category game. Fix category i . Let $z = x(i, L) \in \{0, 1\}$ denote weak-file charging and let $BR_L(z)$ contain the efforts maximizing the police payoff given z . Let $\Phi_L(e)$ be the prosecutor's gain from $z = 1$ rather than $z = 0$, evaluated at the Bayesian posterior induced by effort e . The one-category problem holds the feasible effort interval fixed. Each $BR_L(z)$ is nonempty and compact by continuity. Define

$$e_L = \min BR_L(1), \quad e_H = \max BR_L(0),$$

where the subscripts label the proposed low- and high-effort choices.

At $(z, e) = (1, e_L)$, $e_L \in BR_L(1)$. Condition (iii) gives $\Phi_L(e_L) > 0$, so $z = 1$ is the prosecutor's

best response. At $(z, e) = (0, e_H)$, $e_H \in BR_L(0)$ and $\Phi_L(e_H) < 0$, so $z = 0$ is her best response. In each case, the prosecutor expects the specified effort and updates the probability of guilt by Bayes' rule. Both pairs are therefore mutual best responses. Since $\Phi_L(e)$ is strictly decreasing and $\Phi_L(e_L) > 0 > \Phi_L(e_H)$, $e_H > e_L$.

For the two-category game, let $j \in \{L, H\}$ index these pairs. Specify $z^j = (z_A^j, z_B^j)$ and $\xi^j = (r^j, e_A^j, e_B^j)$, with the focal-category choices $(z_i^L, e_i^L) = (1, e_L)$ and $(z_i^H, e_i^H) = (0, e_H)$. As required by the proposition, $\xi^j \in \mathcal{B}(z^j)$: the police choices maximize their joint payoff under the shared capacity constraint. These choices satisfy the necessary conditions in Lemma 1. Let the commissioner choose r^j and officers choose (e_A^j, e_B^j) there. For each $r \neq r^j$, let officers choose an effort pair in $E_{z^j}(r)$, the set of officer best responses given z^j and r . These strategies make effort optimal at every r and make r^j optimal for the commissioner. The condition $\xi^j \in \mathcal{B}(z^j)$ verifies police incentives across both categories.

In each proposed equilibrium, the prosecutor expects the police to choose ξ^j . Conditional on (i, s) , Bayes' rule gives her posterior probability of guilt $\mu_i(s; \xi^j)$ in equations (A-11)–(A-12). Let $\Phi_i(\xi^j, z^j; \text{lenient})$ denote her gain from $x(i, L) = 1$ rather than $x(i, L) = 0$ at these beliefs. For each $i \in \{A, B\}$, equation (A-14) requires $\Phi_i(\xi^j, z^j; \text{lenient}) \geq 0$ if $z_i^j = 1$ and $\Phi_i(\xi^j, z^j; \text{lenient}) \leq 0$ if $z_i^j = 0$. Thus z^j is optimal, and $x(i, H) = 1$ is required. Write k_j for the resulting continuation strategies and beliefs after $\sigma = \text{lenient}$.

Use the same continuation equilibrium k_{strict} after $\sigma = \text{strict}$ in both proposed PBEs. Let officers choose an effort pair in $E_{(0,0)}(r)$ at every r , and let the commissioner choose $r \in \arg\max_{r \in [0,1]} F_{(0,0)}(r)$, where $F_{(0,0)}$ is the payoff after officers optimize. The prosecutor expects these police strategies and updates the probability of guilt by Bayes' rule. The choice $\sigma = \text{strict}$ requires $x(i, H) = 1$ and $x(i, L) = 0$. Evaluating policy payoffs at the specified strategies and beliefs, if

$$V_P(\text{lenient}; k_j) \geq V_P(\text{strict}; k_{\text{strict}}) \quad \text{for each } j \in \{L, H\},$$

then the prosecutor optimally chooses $\sigma = \text{lenient}$ in each proposed equilibrium. Together with optimal behavior after either σ , these inequalities establish two PBEs. Optimal choices in both categories and the initial preference for $\sigma = \text{lenient}$ are the additional conditions stated in the proposition. $\blacksquare$

Proof of Proposition 5. The proof uses continuity to preserve strict incentives. Let ω be the pre-reform parameter vector and ω' the post-reform vector. Fix either PBE in the hypothesis, and let k_σ denote its continuation strategies and beliefs after σ . Under ω' , retain k_σ for each σ , including officer effort at every r .

For each σ , let $z = (z_A, z_B)$ be the prescribed weak-file charging profile, with $z_i = x(i, L)$, and let $g_i^{z_i}(e)$ be the police return per case net of costs in equation (A-7). Only prosecutor payoff parameters change. Hence $g_i^{z_i}$ and the feasible effort budget are unchanged. Officers' prescribed efforts remain optimal at every r , and the commissioner still maximizes the resulting payoff. The allocation and efforts generate the same signal distribution, so the Bayesian beliefs in k_σ remain valid.

After $\sigma = \text{lenient}$, the prosecutor's gain from $x(i, L) = 1$ rather than $x(i, L) = 0$ satisfies $\Phi(i, L; \text{lenient}) > 0$ when $z_i = 1$ and $\Phi(i, L; \text{lenient}) < 0$ when $z_i = 0$. These gains are continuous in the changed parameters. At ω , $V_P(\text{lenient}; k_{\text{lenient}}) - V_P(\text{strict}; k_{\text{strict}}) > 0$; this difference is also continuous with k_σ fixed. There are finitely many such inequalities across the two PBEs, so all retain their signs in a sufficiently small common neighborhood of ω . The

admissible rules require $x(i, H) = 1$ under either σ and $x(i, L) = 0$ under $\sigma = \text{strict}$. Thus all strategies remain sequentially rational, and both PBEs persist.

If play remains at this equilibrium under ω' , the distribution of case outcomes is unchanged. Let $p_A(r) = r$ and $p_B(r) = 1 - r$ be the category weights, and let $P_i^{z_i}(e_i)$ and $Y_i^{z_i}(e_i)$ be the within-category charging and conviction rates in equations (A-4)–(A-5). The aggregate rates are

$$\bar{P} = \sum_i p_i(r) P_i^{z_i}(e_i), \quad \bar{Y} = \sum_i p_i(r) Y_i^{z_i}(e_i), \quad \Pr(x = 0) = 1 - \bar{P}.$$

The choices (r, e_A, e_B) and z are fixed, as are the guilt and outcome technologies. Every term is therefore unchanged by the reform. The conditional conviction rate $\bar{Y}/\bar{P}$ is unchanged when $\bar{P} > 0$. Appendix A-4 gives two PBEs satisfying the required strict inequalities. ■

Proof of Proposition 6. The proof is by contradiction. Let ω' denote the post-reform parameter vector, let $z = (z_A, z_B)$ specify weak-file charging after $\sigma = \text{lenient}$, with $z_i = x(i, L)$, and let $\xi = (r, e_A, e_B)$ denote the police choices. The set $\mathcal{B}(z; \omega')$ contains the allocations and effort pairs maximizing the joint police payoff given z under ω' . Let $\Phi_i(\xi, z; \omega')$ be the prosecutor's payoff from $x(i, L) = 1$ minus that from $x(i, L) = 0$, evaluated at the Bayesian beliefs induced by ξ after $\sigma = \text{lenient}$.

Suppose first that $\Phi_i(\xi, z; \omega') > 0$ for every z with $z_i = 0$ and every $\xi \in \mathcal{B}(z; \omega')$ with $0 < r < 1$, as in the proposition's first condition. If a continuation equilibrium after $\sigma = \text{lenient}$ prescribed $z_i = 0$, its police choices would satisfy $\xi \in \mathcal{B}(z; \omega')$. At the beliefs induced by ξ , $\Phi_i(\xi, z; \omega') > 0$ makes $z_i = 1$ strictly preferable, contradicting sequential rationality. Thus no such equilibrium can prescribe $z_i = 0$.

Under the second condition, suppose a continuation equilibrium after $\sigma = \text{lenient}$ prescribed $z_i = 1$. Its police choices would satisfy $\xi \in \mathcal{B}(z; \omega')$, where $\Phi_i(\xi, z; \omega') < 0$. The deviation to $z_i = 0$ would be profitable, again contradicting sequential rationality. Thus this condition rules out $z_i = 1$.

By hypothesis, at least one continuation equilibrium after $\sigma = \text{lenient}$ exists under ω' . Its z_i must therefore differ from z_i in at least one of the two pre-reform equilibria. This establishes the claimed change in behavior after $\sigma = \text{lenient}$. Determining the initial choice of σ would additionally require comparing the prosecutor's payoffs across continuation equilibria after both standards. ■

Proof of Implication 1. The proof follows by direct calculation of charging probabilities. Let $z_i = x(i, L) \in \{0, 1\}$ denote weak-file charging and let $h_i(e_i)$ be the probability of $s = H$ at equilibrium effort e_i . Since $x(i, H) = 1$,

$$\Pr(x = 1 \mid i, H) = 1, \quad \Pr(x = 1 \mid i, L) = z_i,$$

and the difference between these conditional charging probabilities is $1 - z_i$. The unconditional charging rate in the category is

$$\Pr(x = 1 \mid i) = h_i(e_i) + z_i[1 - h_i(e_i)].$$

Under unscreened charging, $z_i = 1$ for each i , so $\Pr(x = 1 \mid i) = 1$ and $1 - z_i = 0$. Under screened charging, $z_i = 0$ for each i , so $\Pr(x = 1 \mid i) = h_i(e_i) < 1$ and $1 - z_i = 1$. Under

category-dependent charging, $z_A \neq z_B$, so charging responds to s only in the category with $z_i = 0$. These are the three claims in the implication. $\blacksquare$

A-3 Auxiliary results

A-3.1 Police optimization and effort comparisons

Lemma 2 (Police solutions and sufficient conditions for uniqueness) *For each fixed profile z , continuous primitives and positive capacity give nonempty officer and joint police maximizing sets. If each $g_i^{z_i}$ is strictly concave and C is strictly convex, the joint police solution is unique whenever its allocation is interior. For an interior allocation, concavity of the $g_i^{z_i}$ makes the effort optimality conditions below sufficient as well as necessary.*

Proof of Lemma 2. I prove existence by compactness, uniqueness by strict concavity, and the effort characterization by the first-order conditions. Fix $z = (z_A, z_B)$, where $z_i = x(i, L) \in \{0, 1\}$ denotes weak-file charging. Let $g_i^{z_i}(e)$ be the police return per case net of investigation costs under z_i , as defined in equation (A-7).

For any allocation r , the feasible effort set $\mathcal{F}(r) = \{(e_A, e_B) \in [0, \bar{e}]^2 : re_A + (1 - r)e_B \leq \bar{B}\}$ is nonempty and compact. The continuous officer objective $rg_A^{z_A}(e_A) + (1 - r)g_B^{z_B}(e_B)$ therefore attains a maximum. The joint feasible set of allocations and efforts is

$$\{(r, e_A, e_B) \in [0, 1] \times [0, \bar{e}]^2 : re_A + (1 - r)e_B \leq \bar{B}\},$$

which is also nonempty and compact. Hence the joint police payoff, obtained by subtracting the allocation cost $C(r)$ from the officer objective, attains a maximum. Any joint optimum must use officer-optimal effort at its allocation; otherwise replacing effort improves the joint payoff. It must also maximize the commissioner's payoff after effort is optimized. Conversely, a maximizing commissioner allocation together with officer-optimal effort is a joint optimum. This proves existence and the equivalence of joint and sequential police optimization.

For uniqueness, define total effort in each category by $u_A = re_A$ and $u_B = (1 - r)e_B$. The transformed feasible set is convex:

$$0 \leq r \leq 1, \quad 0 \leq u_A \leq r\bar{e}, \quad 0 \leq u_B \leq (1 - r)\bar{e}, \quad u_A + u_B \leq \bar{B}.$$

The objective becomes

$$rg_A^{z_A}(u_A/r) + (1 - r)g_B^{z_B}(u_B/(1 - r)) - C(r).$$

At $r = 0$ ($r = 1$), define the contribution from category A (B) to be zero, its continuous limiting value on the feasible set. Concavity of each $g_i^{z_i}$ makes each weighted category contribution concave in (r, u_A, u_B) . To see this, per-case effort at a convex combination of two feasible points is a weighted average of their per-case efforts, with weights proportional to their category probabilities. Applying concavity of $g_i^{z_i}$ gives the required inequality. Strict convexity of C then rules out two optima with different r : their convex combination would have a strictly greater

payoff. At a common $r \in (0, 1)$, both category weights are positive, so strict concavity of $g_A^{z_A}$ and $g_B^{z_B}$ rules out two different effort pairs. Thus an optimum with $0 < r < 1$ is unique. For the effort characterization, fix $r \in (0, 1)$ and let $\mu \geq 0$ be the capacity multiplier. Dividing the effort conditions by the positive category weights gives

$$\begin{cases} (g_i^{z_i})'(e_i) = \mu, & 0 < e_i < \bar{e}, \\ (g_i^{z_i})'(0) \leq \mu, & e_i = 0, \\ (g_i^{z_i})'(\bar{e}) \geq \mu, & e_i = \bar{e}, \end{cases} \quad (\text{A-19})$$

together with feasibility and complementary slackness,

$$re_A + (1 - r)e_B \leq \bar{B}, \quad \mu[\bar{B} - re_A - (1 - r)e_B] = 0.$$

These conditions are necessary for optimal effort. When each $g_i^{z_i}$ is concave, they also make each e_i maximize $g_i^{z_i}(e) - \mu e$ on its effort interval. For any feasible alternative effort pair, the gain in the officer objective is therefore at most μ times the increase in total effort, which is nonpositive by feasibility and complementary slackness. This proves sufficiency for the officer problem at the given allocation. For police behavior to be optimal at both stages, the commissioner must also maximize $F_z(r)$ in equation (A-10), anticipating the officers' optimal efforts at each allocation. ■

For a weak-file charging profile z , let $e_i(r)$ denote an officer-optimal effort in category i and $\mu(r) \geq 0$ the corresponding capacity multiplier. When the optimized commissioner value $F_z(r)$ is differentiable and the envelope formula applies, its derivative is

$$F'_z(r) = g_A^{z_A}(e_A(r)) - g_B^{z_B}(e_B(r)) - \mu(r)[e_A(r) - e_B(r)] - C'(r). \quad (\text{A-20})$$

An interior optimum has $F'_z(r) = 0$. A boundary maximum at zero has $F'_z(0) \leq 0$ and one at unity has $F'_z(1) \geq 0$, where one-sided derivatives exist. These remain necessary rather than sufficient without concavity.

At a boundary allocation, effort in the category with zero weight has no effect on payoffs. Its charging information sets are not reached. A boundary PBE construction must specify off-path beliefs there and check that they support the proposed lenient charging decisions. Bayes' rule fixes beliefs only in the category reached with positive probability. The main regime results avoid this extra freedom by stating their interior scope. In particular, an interior minimum of C alone does not rule out a boundary optimum.

Lemma 3 (The effort incentive from screening) *Fix a category and a common feasible effort interval. If $v_i > 0$ and $h'_i(e) > 0$, every optimal effort under screening is weakly greater than every optimal effort under broad charging. The inequality is strict if both compared optima are interior. This comparison applies to the two-category model when the unconstrained category optima are feasible under both profiles. It does not in general rank category efforts after the common capacity constraint changes its shadow price.*

Proof of Lemma 3. I prove the weak ordering by comparing optimality inequalities, then prove strictness by contradiction. Fix category i and the common feasible effort interval in the

hypothesis. Let $z_i = x(i, L) \in \{0, 1\}$ denote weak-file charging, $h_i(e)$ the probability of $s = H$, and $t_{is}(e)$ the joint probability of signal s and $y = 1$ when $x(i, s) = 1$, as in equations (A-1)–(A-3). The net police returns under $z_i = 0$ and $z_i = 1$ are

$$g_i^0(e) = v_i h_i(e) + w_i t_{iH}(e) - c(e), \quad g_i^1(e) = v_i + w_i [t_{iH}(e) + t_{iL}(e)] - c(e).$$

The superscript records z_i . Define the difference in returns by

$$d_i(e) = g_i^0(e) - g_i^1(e) = -v_i [1 - h_i(e)] - w_i t_{iL}(e).$$

Differentiating $t_{iL}(e)$, the joint probability of $s = L$ and $y = 1$ when $z_i = 1$, gives

$$t'_{iL}(e) = -\pi_i \alpha'_i(e) q(1, L) - (1 - \pi_i) \beta'_i(e) q(0, L) \leq 0.$$

Since $v_i > 0$, $w_i \geq 0$, and $h'_i(e) > 0$, it follows that

$$d'_i(e) = v_i h'_i(e) - w_i t'_{iL}(e) > 0. \quad (\text{A-21})$$

Let e_0 be any maximizer of g_i^0 and e_1 any maximizer of g_i^1 on the common interval. Optimality implies

$$g_i^0(e_0) \geq g_i^0(e_1), \quad g_i^1(e_1) \geq g_i^1(e_0).$$

Adding yields $d_i(e_0) \geq d_i(e_1)$. Because d_i is strictly increasing, $e_0 \geq e_1$. If both optima were interior and equal, the derivatives of both net returns would be zero at the same effort. Their difference would then have derivative zero, contradicting equation (A-21). Thus $e_0 > e_1$ when both optima are interior. The comparison applies on the common feasible interval stated in the lemma; different capacity prices in the full police problem require a separate comparison. ■

A-3.2 Charging incentives and policy comparisons

Proposition 7 (Conditional comparative statics) *The following comparisons hold at the specified behavior and beliefs.*

1. *The weak-file charging gain is weakly increasing in b and Δ_E , and weakly decreasing in the acquittal cost a and wrongful-conviction cost λ , holding other payoff terms fixed.*
2. *Let $K = (b + a)(q_{1L} - q_{0L}) + \lambda q_{0L}$. If $K > 0$, weak-file charging incentives decrease strictly with effort whenever*

$$\frac{\alpha'_i}{1 - \alpha_i} > \frac{\beta'_i}{1 - \beta_i} \quad \text{and} \quad \frac{d\Delta_E}{de} \leq 0.$$

The signal inequality follows from the maintained $\alpha_i > \beta_i$ and $\alpha'_i > \beta'_i \geq 0$.

3. *For fixed continuation strategies and beliefs k_L after leniency and k_S after strictness, let $\Delta V_P = V_P(\text{lenient}; k_L) - V_P(\text{strict}; k_S)$. For $j \in \{L, S\}$, write $\bar{P}_j$, $\bar{Y}_j$, and $\bar{W}_j$ for the aggregate charging, conviction, and wrongful-conviction rates, and R_j for expected*

retention. Holding strategies, outcome technologies, and the electoral rule fixed,

$$\begin{aligned}\frac{\partial \Delta V_P}{\partial b} &= \bar{Y}_L - \bar{Y}_S, & \frac{\partial \Delta V_P}{\partial a} &= -[(\bar{P}_L - \bar{Y}_L) - (\bar{P}_S - \bar{Y}_S)], \\ \frac{\partial \Delta V_P}{\partial \lambda} &= -(\bar{W}_L - \bar{W}_S), & \frac{\partial \Delta V_P}{\partial \rho} &= R_L - R_S.\end{aligned}$$

These are conditional incentive comparisons. Extending them to equilibrium regime regions requires the same continuations to remain sequentially rational and the initial policy comparison to retain the required sign.

Proof of Proposition 7. I prove the claims by differentiating the charging gain and the difference in policy payoffs. For category i , let $\mu_i(L)$ be the posterior probability of guilt conditional on $s = L$ and let $Q_i(L) = \mu_i(L)q(1, L) + [1 - \mu_i(L)]q(0, L)$ be the probability of $y = 1$ if $x(i, L) = 1$. Write Φ_i for the payoff from $x(i, L) = 1$ minus that from $x(i, L) = 0$, and Δ_E for its electoral component. Equation (A-13) gives

$$\Phi_i = (b + a)Q_i(L) - a - \lambda[1 - \mu_i(L)]q(0, L) + \Delta_E.$$

Holding beliefs and the other payoff terms fixed,

$$\frac{\partial \Phi_i}{\partial b} = Q_i(L), \quad \frac{\partial \Phi_i}{\partial a} = -(1 - Q_i(L)), \quad \frac{\partial \Phi_i}{\partial \lambda} = -(1 - \mu_i(L))q(0, L), \quad \frac{\partial \Phi_i}{\partial \Delta_E} = 1.$$

The bounds $0 \leq Q_i(L), \mu_i(L), q(0, L) \leq 1$ give the signs in part (i).

For part (ii), let e be per-case effort in category i and let $h_i(e) = \pi_i \alpha_i(e) + (1 - \pi_i) \beta_i(e)$ be the probability of $s = H$. The posterior conditional on $s = L$ is $\mu_i(L) = \pi_i[1 - \alpha_i(e)]/[1 - h_i(e)]$. Differentiating gives

$$\frac{d\mu_i(L)}{de} = \frac{\pi_i(1 - \pi_i)\{[1 - \alpha_i(e)]\beta'_i(e) - [1 - \beta_i(e)]\alpha'_i(e)\}}{[1 - h_i(e)]^2}. \quad (\text{A-22})$$

Since $0 < h_i(e) < 1$, the denominator is positive. The signal inequality in the proposition makes the numerator negative. Define $K = (b + a)[q(1, L) - q(0, L)] + \lambda q(0, L)$, the coefficient on $\mu_i(L)$ in the charging payoff. Then

$$\Phi_i = (b + a)q(0, L) - a - \lambda q(0, L) + K\mu_i(L) + \Delta_E,$$

so $d\Phi_i/de = K d\mu_i(L)/de + d\Delta_E/de < 0$ when $K > 0$ and $d\Delta_E/de \leq 0$, as assumed.

For part (iii), let k_L and k_S be fixed continuation strategies and beliefs after $\sigma = \text{lenient}$ and $\sigma = \text{strict}$, respectively. Write $\bar{P}_j$, $\bar{Y}_j$, and $\bar{W}_j$ for their aggregate charging, conviction, and wrongful-conviction rates, and R_j for their expected retention probabilities, where $j \in \{L, S\}$. Define the prosecutor's payoff difference by $\Delta V_P = V_P(\text{lenient}; k_L) - V_P(\text{strict}; k_S)$. Equation (A-15) gives

$$\begin{aligned}\Delta V_P &= b(\bar{Y}_L - \bar{Y}_S) - a[(\bar{P}_L - \bar{Y}_L) - (\bar{P}_S - \bar{Y}_S)] \\ &\quad - \lambda(\bar{W}_L - \bar{W}_S) + \rho(R_L - R_S).\end{aligned}$$

Holding the strategies, outcome technologies, and electoral rule fixed gives

$$\begin{aligned}\frac{\partial \Delta V_P}{\partial b} &= \bar{Y}_L - \bar{Y}_S, & \frac{\partial \Delta V_P}{\partial a} &= -[(\bar{P}_L - \bar{Y}_L) - (\bar{P}_S - \bar{Y}_S)], \\ \frac{\partial \Delta V_P}{\partial \lambda} &= -(\bar{W}_L - \bar{W}_S), & \frac{\partial \Delta V_P}{\partial \rho} &= R_L - R_S.\end{aligned}$$

These are the claimed conditional derivatives. If a parameter change alters continuation behavior, the total change in policy payoffs also includes that response. ■

Two further distinctions follow directly from these expressions. Identical category primitives and identical effort and electoral increments imply identical weak-file gains, so strictly opposite charging incentives cannot occur at that symmetric assessment. Symmetric primitives alone need not force symmetric behavior when coordination permits multiple assessments. Also, the charging return to investigation is $v_i[\pi_i \alpha'_i + (1 - \pi_i) \beta'_i]$. A larger level gap $\alpha_i - \beta_i$ is not by itself a larger marginal charging return. Changes in courtroom sensitivity affect conviction payoffs and beliefs-based charging incentives, but do not enter this charging derivative.

A-3.3 Uniqueness and continuation selection

Let $\mathcal{C}_L$ be the nonempty set of lenient continuation equilibria. Each uses an interior global police optimum $\xi \in \mathcal{B}(z)$ for a weak-file charging profile z , officer optima after every allocation, and Bayesian beliefs satisfying equation (A-14). When discussing uniqueness, assume all relevant police maximizers are interior. Fix a strict continuation equilibrium k_S , write $V_S = V_P(\text{strict}; k_S)$ for its prosecutor payoff, and let

$$\mathcal{V}_L = \{V_P(\text{lenient}; k) : k \in \mathcal{C}_L\}.$$

Proposition 8 (Uniqueness and continuation selection) *Under the maintained interior scope, the following statements hold.*

1. *Suppose a profile $\hat{z}$ has the following property: at every candidate (z, ξ) with $\xi \in \mathcal{B}(z)$, $\Phi_i(\xi, z; \text{lenient}) > 0$ if $\hat{z}_i = 1$ and $\Phi_i(\xi, z; \text{lenient}) < 0$ if $\hat{z}_i = 0$. Then every lenient continuation has profile $\hat{z}$. If $\mathcal{B}(\hat{z})$ is a singleton, its on-path outcome is unique.*
2. *If $V_S > \sup \mathcal{V}_L$, strictness is chosen in every full PBE using that strict continuation. If $\inf \mathcal{V}_L > V_S$, leniency is chosen in every such PBE. Uniqueness of behavior additionally requires uniqueness of the continuation outcome chosen on path.*
3. *If there are $k_-, k_+ \in \mathcal{C}_L$ with $V_P(\text{lenient}; k_-) < V_S < V_P(\text{lenient}; k_+)$, there is a full PBE choosing strictness and another choosing leniency, despite the strict payoff comparisons.*

Uniqueness here concerns on-path behavior; strategies at unreachable histories may still differ.

Proof of Proposition 8. I prove the first claim by ruling out alternative charging profiles, the second by comparing policy payoffs, and the third by constructing two PBEs. Let $z = (z_A, z_B)$ denote weak-file charging, with $z_i = x(i, L)$, let $\xi = (r, e_A, e_B)$ denote police choices, and let $\mathcal{B}(z)$

contain the globally optimal police choices given z . The gain $\Phi_i(\xi, z; \text{lenient})$ from $x(i, L) = 1$ rather than $x(i, L) = 0$ is evaluated at the Bayesian beliefs induced by ξ after $\sigma = \text{lenient}$. For part (i), let $\hat{z}$ be the profile in the hypothesis. Suppose a continuation equilibrium after $\sigma = \text{lenient}$ used $z \neq \hat{z}$. Then $z_i \neq \hat{z}_i$ for some i . Its police choices satisfy $\xi \in \mathcal{B}(z)$, where the assumed sign of Φ_i makes $\hat{z}_i$ strictly preferable to z_i . This contradicts sequential rationality. Conversely, select $\xi = (r, e_A, e_B) \in \mathcal{B}(\hat{z})$. Let the commissioner choose r and officers choose (e_A, e_B) there. At every other allocation, let officers maximize their payoff subject to the effort bounds and capacity constraint. These police strategies are optimal given $\hat{z}$. The prosecutor expects the police to choose ξ . Conditional on (i, s) , Bayes' rule gives the posterior probability of guilt $\mu_i(s; \xi)$ in equations (A-11)–(A-12). At these beliefs, the assumed signs of Φ_i make $\hat{z}$ optimal. If $\mathcal{B}(\hat{z})$ has one element, (r, e_A, e_B) is unique; Bayes' rule then fixes beliefs at all reached charging information sets. This proves uniqueness of the continuation outcome after $\sigma = \text{lenient}$.

For part (ii), let $\mathcal{C}_L$ contain the continuation equilibria after $\sigma = \text{lenient}$ and let $\mathcal{V}_L = \{V_P(\text{lenient}; k) : k \in \mathcal{C}_L\}$ be their prosecutor payoffs. Fix a continuation equilibrium k_S after $\sigma = \text{strict}$ and define $V_S = V_P(\text{strict}; k_S)$. If $V_S > \sup \mathcal{V}_L$, the prosecutor chooses $\sigma = \text{strict}$ in every PBE using k_S . If $\inf \mathcal{V}_L > V_S$, she chooses $\sigma = \text{lenient}$ in every such PBE. A unique σ yields a unique outcome only if its prescribed continuation outcome is also unique.

For part (iii), let $k_-, k_+ \in \mathcal{C}_L$ satisfy $V_P(\text{lenient}; k_-) < V_S < V_P(\text{lenient}; k_+)$. First, prescribe k_- after $\sigma = \text{lenient}$ and k_S after $\sigma = \text{strict}$. Their strategies are sequentially rational and their beliefs satisfy Bayes' rule. Since $V_P(\text{lenient}; k_-) < V_S$, the prosecutor optimally chooses $\sigma = \text{strict}$, completing one PBE. Next, prescribe k_+ after $\sigma = \text{lenient}$ and the same k_S after $\sigma = \text{strict}$. Since $V_P(\text{lenient}; k_+) > V_S$, she optimally chooses $\sigma = \text{lenient}$, completing a second PBE. The two PBEs have different σ despite strict payoff comparisons. ■

A-3.4 Reallocation of enforcement

Proposition 9 (Screening and police reallocation) *Consider fixed charging profiles z^L, z^S under lenient and strict screening, with the same police primitives and capacity. Let $F_L(r)$ and $F_S(r)$ be the optimized commissioner objectives in equation (A-10). Suppose each has a unique maximizing allocation. If $F_S(r) - F_L(r)$ is weakly increasing in r , then $r_S \geq r_L$. When the objectives are differentiable, a sufficient condition is*

$$\begin{aligned} g_A^{z_A^S}(e_A^S(r)) - g_B^{z_B^S}(e_B^S(r)) - \mu_S(r)[e_A^S(r) - e_B^S(r)] \\ \geq g_A^{z_A^L}(e_A^L(r)) - g_B^{z_B^L}(e_B^L(r)) - \mu_L(r)[e_A^L(r) - e_B^L(r)] \end{aligned}$$

throughout the allocation interval. Capacity terms vanish where both budgets are slack. The comparison still includes conviction returns and effort costs.

Proof of Proposition 9. The allocation ordering is proved by contradiction; the derivative condition follows by comparing marginal payoffs. For $j \in \{L, S\}$, let $z^j = (z_A^j, z_B^j)$ be the fixed weak-file charging profile, with L denoting $\sigma = \text{lenient}$ and S denoting $\sigma = \text{strict}$. Let $F_j(r)$ be the commissioner's payoff after officers optimize their efforts subject to capacity, and let r_j be its unique maximizing allocation. These are F_{z_j} and its maximizer in equation (A-10).

Suppose $r_S < r_L$. Since each allocation is optimal for its own objective,

$$F_S(r_S) \geq F_S(r_L), \quad F_L(r_L) \geq F_L(r_S).$$

Adding gives

$$[F_S - F_L](r_S) \geq [F_S - F_L](r_L).$$

The hypothesis that $F_S - F_L$ is weakly increasing gives the reverse inequality when $r_S < r_L$. Equality must therefore hold. Each of the two original optimality inequalities has a nonnegative gap, so both gaps must be zero. This makes two different allocations maximize each objective, contradicting uniqueness. Hence $r_S \geq r_L$.

To verify the stated sufficient condition, let $e_i^j(r)$ be optimal per-case officer effort in category i under profile z^j , let $\mu_j(r) \geq 0$ be its capacity multiplier, and let $g_i^{z_i^j}(e)$ be the category's net police return. Where the optimized values are differentiable and the envelope formula in equation (A-20) applies,

$$F_j'(r) = g_A^{z_A^j}(e_A^j(r)) - g_B^{z_B^j}(e_B^j(r)) - \mu_j(r)[e_A^j(r) - e_B^j(r)] - C'(r).$$

Subtracting these expressions cancels $C'(r)$. The inequality in the proposition is exactly $F_S'(r) - F_L'(r) \geq 0$. It makes $F_S - F_L$ weakly increasing and therefore establishes $r_S \geq r_L$. With slack capacity, $\mu_j(r) = 0$; the net returns still include conviction benefits and investigation costs. ■

A-4 Multiplicity under a lenient standard

Under a lenient standard, broad charging with zero investigative effort and screened charging with positive effort can both be equilibria of the full game. Consider identical categories with

$$\begin{aligned}\pi_i &= 0.6, & \alpha_i(e) &= 0.2 + 0.6e, & \beta_i(e) &= 0.1, & \bar{e} &= 1, & \bar{B} &= 0.8, \\ v_i &= 1, & w_i &= 0, & c(e) &= 0.3e^2, & C(r) &= 2(r - \frac{1}{2})^2, \\ q_{1H} &= 0.9, & q_{0H} &= 0.2, & q_{1L} &= 0.5, & q_{0L} &= 0.1, \\ b &= 1, & a &= 0.43, & \lambda &= 0.1, & \rho &= 1.\end{aligned}$$

Let $\Gamma = 0.4$ under leniency and $\Gamma = 0.2$ under strictness, independent of case outcomes, so $\Delta_E = 0$.

Police returns are $g_i^1(e) = 1 - 0.3e^2$ under broad charging and $g_i^0(e) = 0.16 + 0.36e - 0.3e^2$ under screening. Their unique optima are $e = 0$ and $e = 0.6$, respectively. Under either uniform charging rule, these efforts satisfy the budget for every r . Identical optimized returns across categories imply $r = 1/2$. These effort choices also specify optimal officer behavior after every other allocation.

At $e = 0$ and $e = 0.6$, Bayes' rule gives $\mu_i(L) = 4/7$ and $11/26$, respectively. The weak-file charging payoff is

$$\Phi_i(L) = 1.43[0.1 + 0.4\mu_i(L)] - 0.43 - 0.01[1 - \mu_i(L)].$$

It equals $0.035571... > 0$ under broad charging and $-0.050769... < 0$ under screening. Strong-file gains are positive in both cases. Thus both lenient continuations satisfy charging incentives. Under strictness, screening is mandatory and induces effort $e = 0.6$.

	Lenient, broad	Lenient, screened	Strict
r	0.5	0.5	0.5
$e_A = e_B$	0	0.6	0.6
$\bar{P}$	1	0.376	0.376
$\bar{Y}$	0.392	0.3104	0.3104
$\bar{P} - \bar{Y}$	0.608	0.0656	0.0656
$\bar{W}$	0.044	0.008	0.008
V_P	0.52616	0.681392	0.481392

Table A-2: Two equilibria under leniency, with the strict continuation used for the policy comparison.

Both lenient payoffs exceed the strict payoff in Table A-2, calculated from equation (A-15). Prescribing the strict continuation after a deviation therefore supports two full PBEs with the same lenient standard. The strict payoff inequalities ensure that sufficiently small changes in a or λ preserve both equilibria and their outcome rates, as in Proposition 5.

A-5 Modeling choices

The prosecutor’s publicly observed standard restricts subsequent charging decisions. Corroboration requirements and supervisory approval rules provide institutional counterparts: they constrain which cases may be filed while leaving discretion within those bounds (Barkow 2019, Davis 2007). The binary specification captures the difference between requiring a strong file and permitting weak files. A lenient policy can produce screened behavior when declining weak files is optimal within its larger admissible set.

Police investment is not observed at the charging stage. The prosecutor evaluates a file using the equilibrium distribution of evidence, updating on its category and signal. The commissioner’s allocation is likewise not an additional charging-stage observation in the stated information structure. If that allocation were publicly observed by the prosecutor, charging strategies could depend on r ; one would then need a continuation equilibrium following every observed allocation, and the fixed-profile commissioner optimization used here would change.

The commissioner sets priorities before officers choose effort. They share an objective, so solving their two stages is equivalent to finding a global joint police optimum for a given charging profile. Sequential strategies nevertheless require an effort prescription after every allocation. The shared budget couples the categories, while $C(r)$ makes departures from the preferred case mix costly. These restrictions have different roles, and neither implies equal case shares.

The court and electoral evaluation are specified rather than solved as strategic choices. The court’s probability $q(\theta, s)$ captures how the state and evidence affect conviction. The electoral increment is the conditional payoff difference under the chosen evaluation rule; it cannot be chosen independently of the rule used to calculate the value of a policy. Generalizing the voter or allowing the prosecutor to choose a portfolio-wide rule under nonlinear electoral payoffs would require additional analysis.

Finally, I interpret screening as applying to cases in which trial viability constrains charging. Cases resolving by plea regardless of the standard contribute to baseline outcomes. Police produce evidence before prosecutorial review; prosecutor-directed requests for further investigation are outside the timing analyzed here.